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Deep Learning

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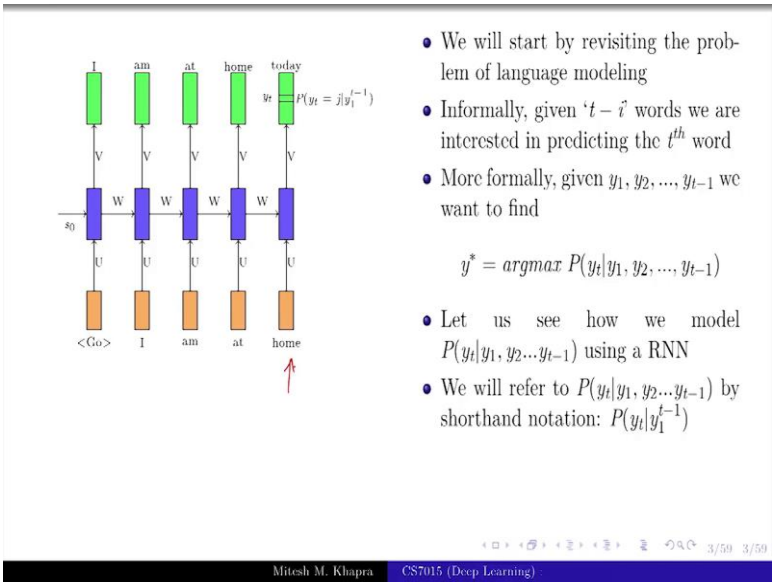
Indian Institute of Technology, Madras

Lecture – 15

Encoder Decoder Models, Attention and Mechanism

And in this lecture, we are going to talk about Encoder Decoder Models and Attention Mechanism. So, this is a very interesting lecture at least interesting to me because, this is very put all these pieces that we have learnt so far right. We have learnt 3 types of networks; feed-forward networks, recurrent neural networks and convolutional neural networks and we have seen independent applications of each of these word to vec and image classification and so on. Now today, what we are going to see is how do we do different combinations of these networks and come up with a wide range of applications like apply them to a wide range of applications ok. So, let me start by an introduction to encoder decoder models and then we do various applications of encoder decoder models.

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So, what we are going to do is we will start by revisiting the problem of language modeling. So, the problem of language modeling was that you are given some t - 1 words or characters and you want to predict the tth word or character right, this is like auto complete in short right, whenever we are typing something, you have type 4 words, you want to predict the fifth word. Or you have typed 4 characters and you want to predict the fifth character ok.

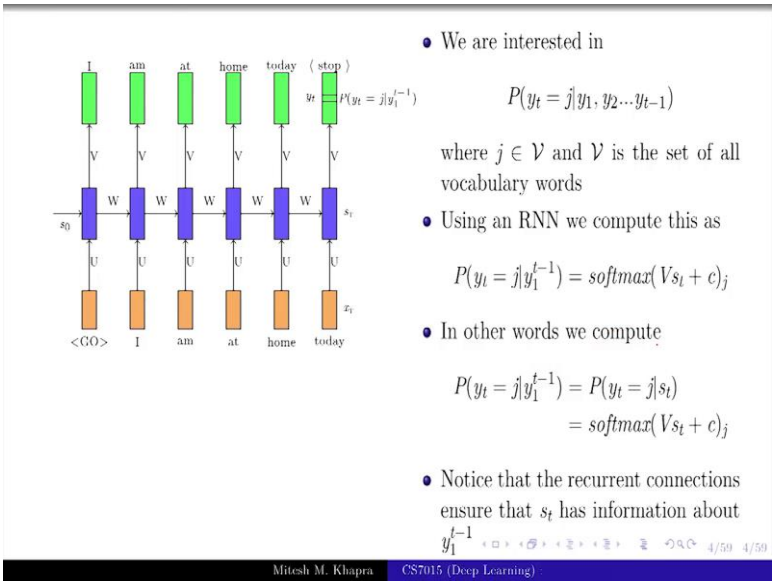
So, more formerly this is what we are interested in. How many of you get this equation this expression? So, we are given a sequence of t - 1 words and you want to find out, what the value of y would be at time step t and we want to find out that value which maximizes this property. That is what this argmax equation means and now we will try to see, how to model this using a RNN.

So, let us see. We are going to start with go that is that, we want to start generating a sentence and then we will produce the first word which is I, ok, and what is it that we are predicting at this point, what is the network supposed to predict, what is the output supposed to predict actually?

It is supposed to predict a dash over the vocabulary, a broadly distribution over the vocabulary right. So, this is what is happening. We will of course come back to this on the next few slides, but you have say words W1, W2 up to W V in your vocabulary. At every time step, you want to find a distribution over these words and then pick the word, which had the maximum probability at that time step right; that is exactly, what this quantity is that is what we want the RNN in to model and then we want to keep doing this till, we reach the end of the sentence ok. So, that is the language modeling problem and as we had made a case for it earlier, the word produce the time step t depends on a few previous words. How does a recurrent neural network ensure that? At any time step, I am going to give it only 1 word as the input.

So, how does that ensures that it depends on all the previous words also? Through the recurrent connections and the gate and sorry it is not the gate the state st, ok, fine.

(Refer Slide Time: 03:03)



So, we will see this of course, in more detail and we will write down the model equations and what is happening. So, we are interested in this quantity, which is the probability of the word at the time at the t th time step, where this j belongs to vocabulary V and see a vocabulary of 10k words or 20k words for English it is actually much higher, but say you are considering only 10k to 20k words, then we want to predict a distribution over this vocabulary. So, using an RNN what are you going to do at the output layer is the following, is this correct? How many if you understand this equation? Not many why? What does this equation compute? First of all softmax, softmax means?

Student: Probability distribution.

Probability distribution. Ok. What does it take as input at every time step? The state right, what does it do with the state? A linear transformation right and then a bias ok. So, what is this quantity, scalar vector matrix, vector of size?

Students: (Refer Time: 04:00).

The (Refer Time: 04:01). What is the gth element of the that?

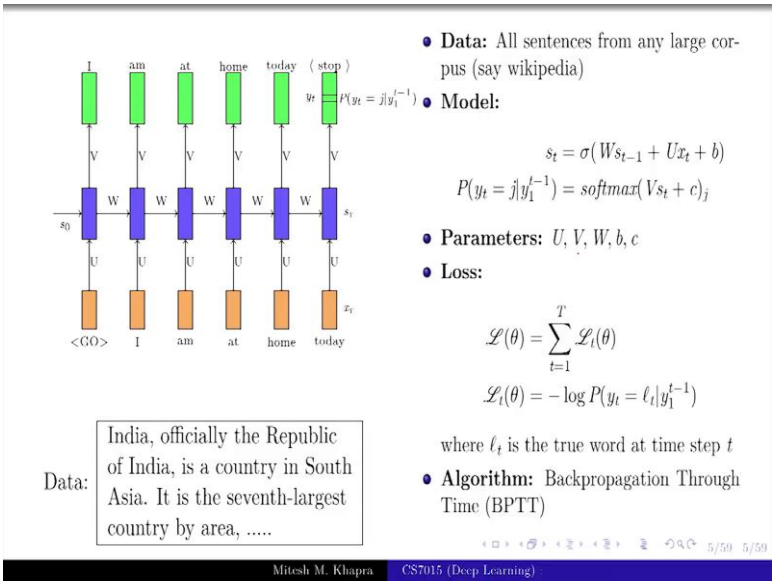
Students: Probability of the gth word.

The probability of the gth word right. So, I just have to explain it in that many words, everyone gets it, now, everyone gets it? If you do not get it, you will not understand the rest of the lecture. I am very serious everyone gets it?

So in other words, what we do this entire y_1 to y_{t-1}, which we were conditioning on we are just using st as a surrogate for that and that is fair because, st has actually captured all the previous information that we had, now just using st as a state which captures everything that happened so far.

So, that is actually how we are modeling this and the recurrent connections ensures that st captures everything which has happened so far.

(Refer Slide Time: 04:43)



So now, let us look at the 5 things that we have in a typical supervised machine learning set up which are those? Data, model.

Students: Parameters.

Parameters.

Students: Objective function.

Objective function.

Student: Cross (Refer Time: 04:54).

Very good, no someone said objective function and then loss function, learning algorithm right ok. So, whats the model here?

Students: (Refer Time: 05:01).

You know, what you are trying to model which is a property distribution, what is the actual? So, here y is the probability distribution and your x is the input given to you, can you tell me what's and we have already set always said in this course that, whatever be the y whatever the be the x, we are interested in this function x sorry function f and we should be actually expressively, we able to write this function. So, what is the function here? Can you actually write down the set of equation? Just think of what the output is, how you are going to reach the output given this network? What is yt going to be, what's the equation for yt?

And then try to go back all the way back to xt. So, yt depends on something that something might depend on xt. So, how do you go all the way back? Right, that is the thing, which I expect you to do. Ok. How many of you get it now? Please raise your hands ok. So, let us see, at every time step, what am I interested in predicting?

Students: Probability distribution.

A probability distribution; that means, I will have to compute which function?

Students: Softmax.

Softmax. So, the green vector is what I am going to focus on. So, what's the equation for the green vector? Is this fine? Now what does this contain apart from the parameters st? How do I get st? Is it fine? You can write, now you have written this output y as a function of x because, x appears here or other yt as a function of xt is not it is straight forward right once, I show you the answer is should be how many of you get it now? Please raise your hands high up above, okay. What are the parameters? B and c right.

So, these are the parameters, what's the objective function? Cross entropy or dash of cross entropies.

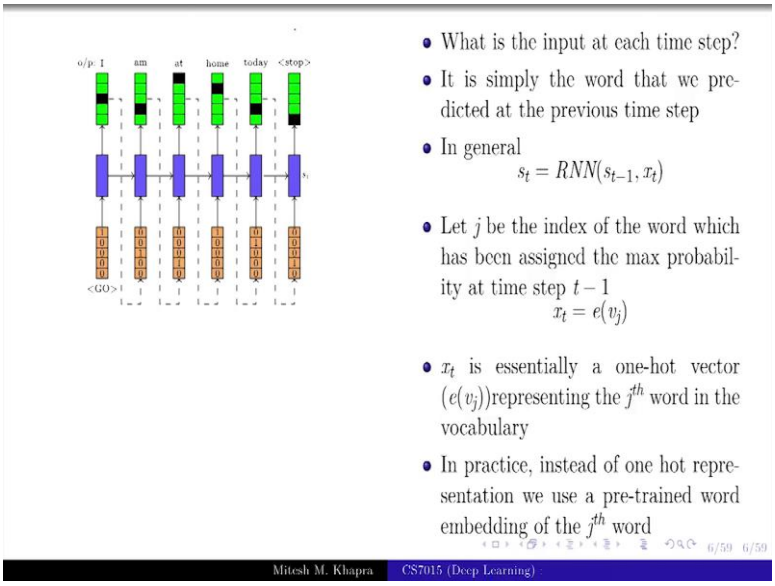
Students: Sum of cross entropies.

Sum of cross entropies right. So, the loss is going to be over all the time steps, at every time step is the cross entropy loss right, everyone gets this? Ok, what's the learning algorithm? Back propagation.

Students: True time.

True time, fine. So, that is what it is going to be right everyone is clear? So, you can see that we have written the final output as a function of the input right and this is end to end trainable; that means, the gradients can flow modulo this vanishing exploding radiant problem and we have a way of handling that, we can replace RNS by LSTMs that is all right . So, that is what it is now, this, just make sure you understand this properly. So, that we are going to do various instantiations of this for different problems ok.

(Refer Slide Time: 07:31)



Now here is 1 question, we all smartly wrote this xt, but why is the input at every time step? When I am predicting home, the input was at, but how did I get at? That is what I dash at the previous time step, predicted at the previous time step right.

So, this is what the input looks like. So, at time step 1 I predicted I as the output at the next time step, I am going to feed that has the input, does it make sense? So, just see if you are doing auto complete, you would select that I am fine with the word I at this time step. So, it is going to take that as the input and then try to predict the next word that is

what exactly is happening here and now you are predicted am at the next time step you are going to feed am as the input and continue this chain throughout ok.

So, the input at every time step is going to be the word that you have predicted the previous time step and I am just going to represent it by a 1 hot vector right. It is the index of the jth word only that could be hot everything else would be 0 and all of you are fine with this. No so at training time this is the inference time, at training time we will have the real inputs no, that is at inference time, at training time, we will just use that through because, training time you know what the inputs are right, you know the true sentence you have the Wikipedia sentence right and you know what the true sentence is going to be.

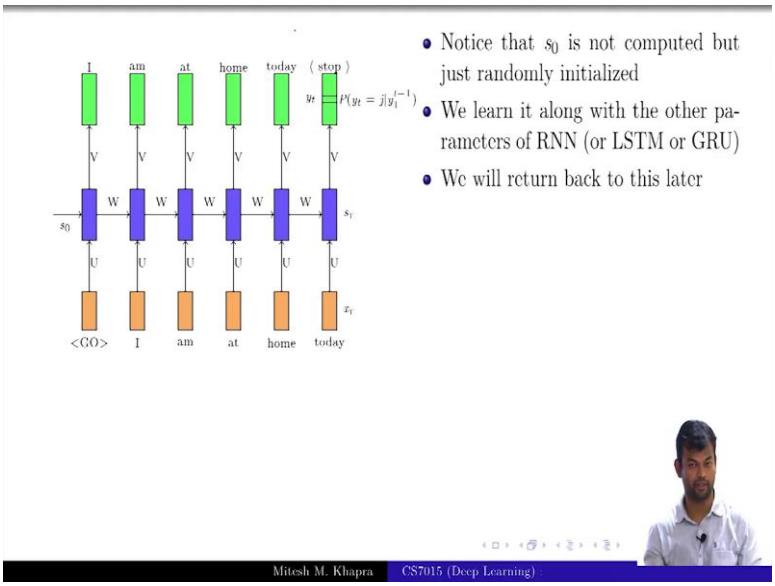
What I am talking about, how will you generated at test time. At training time, you know all these things right. No about training time, how will you do that? You will know what the next input is right. So now, ok. So, I said that the input is going to be a 1 hot vector. is everyone fine with that? 1 hot vectors are ok, what else could you use?

Students: Word representation.

The word representation for that right. So, assume that you have already done the word to vec assignment and you have completed all the word representations and you have them with you now. But instead of feeding the one hot representation of the input, you can just feed the word representation of it does that make sense? One hot representation is just one of the many representations possible for the world. So, why just do that, you could do s v d, you could do one word vec or whatever you want right. So, that is in practice, what we will feed is the word to vec representation.

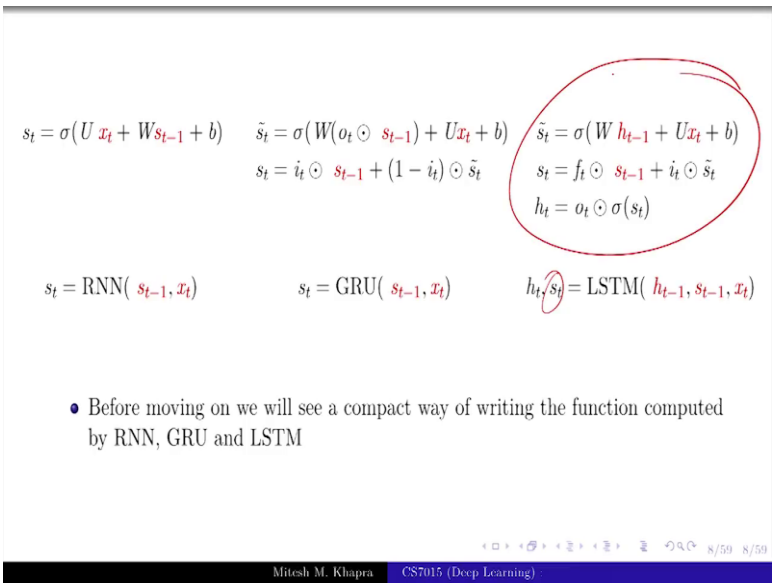
So, everyone gets this what is happening at every time step?

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Now, one more thing, that you need to notice that s_0; which is the input at time step 1, the previous; so, s_{1-1}. So, that we do not know what it is. So, we just keep it as a parameter, we say that s_0 is also weight vector and you are going to learn it, along with all the other parameters in the network. Does not make sense, because you do not know, what s_0 means is a semantics of it is not clear like, what was generated at the 0th time step, we do not really know right. So, will just make it a learnable parameter and that would be trained along with all the other parameters of the network.

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So, before we move on what we are going to do is, we are going to see a very compact representations for RNNs, GRUs and LSTMs. So, remember RNN is the following equation RNN is defined by the following equation. The s_t is a recursive function of s_{t-1} and x_t right. So, I am just going to write it as that s_t is equal to RNN of s_{t-1}, x_t, instead of writing all these parameters and sigma's and all that, I am just going to write it compactly as this, now this is what, what is this? GRU. So, how may going to write it as?

Students: GRU.

GRU of?

Students: S_{t-1} x_t.

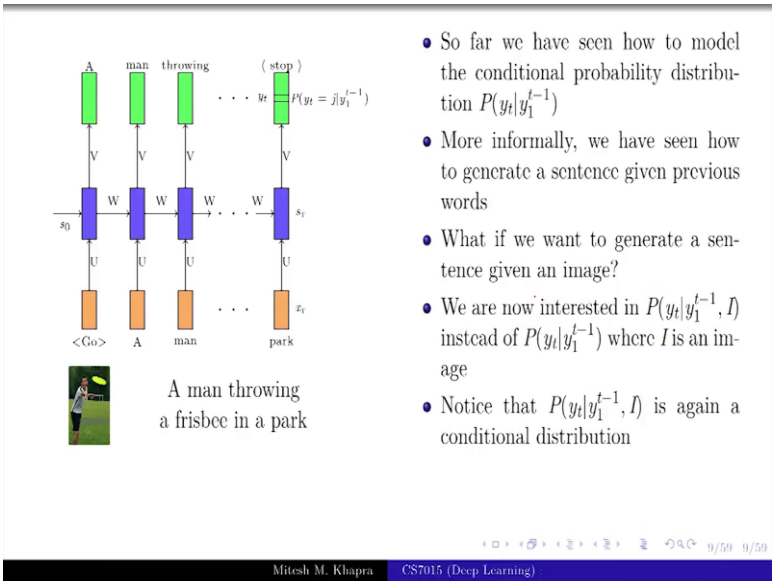
S_{t-1}, x_t.

What is this?

Students: LSTM.

LSTM, how may going to write it? LSTM of when the output of the LSTM is both h_{t-1} and s_{t-1}right fine. So, in some sometimes I will just say st, sometimes I will say both ht minus ht and st as per whatever I needed right. So, this is I am not going to write these equations and parameters again, I will just say that LSTM of this, assume that is a function which does this calculation and gives you back ok? Ok.

(Refer Slide Time: 11:22)



So, far what you have done is we have seen, how to model the conditional probability distribution given the previous t minus 1 words. Now, let me give you a different application right, what if we want to generate a sentence given an image. So, this is what I am interested in doing, I am giving an image and I want to generate a sentence, can we just think of it formally, what is it that you want to do? So, we saw that in this case formally, we were interested in this conditional distribution. In this case, what is it that we are formally interest in?

If I were to write it as something formal, what would I write it as? Ok I will give you a hint, what kind of a distribution is this? A conditional distribution right, given the previous sequences previous sequence of words generate the tth word, now in this situation can you stated an similar words? Given the.

Students: Image.

Image generate the.

Students: Sentence.

Sentence or given the image and the description that are generated so far, because I am going to write the description, 1 word at a time given, the image and the description that have written so far, generate the next word in the mission. So, what kind of conditional distribution is that? $P(y_t)$ given.

Students: Y 1 to t minus 1.

Y1 to t minus 1?

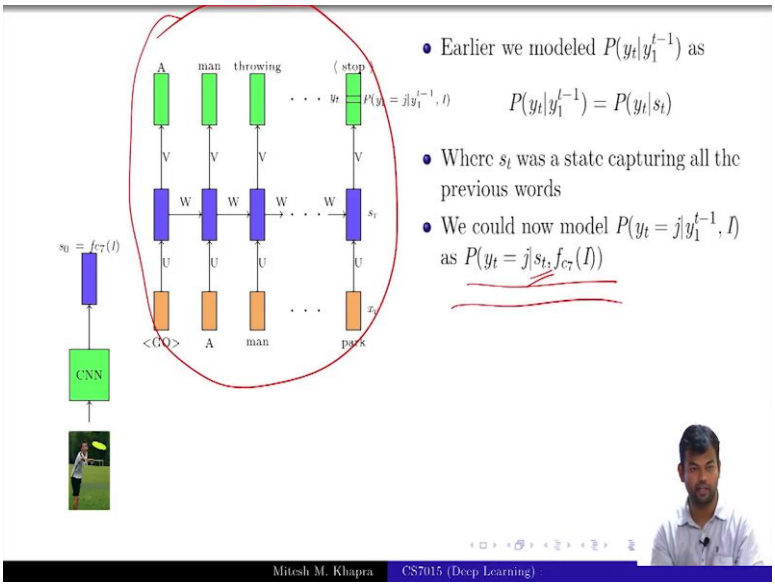
Students: Comma.

Comma?

Students: Image.

Image. Does that make sense? Everyone gets that. Ok? So what. So, this is what we want right. So here, now we are interested in this quantity as a post to this quantity. Does that make sense? Ok and this is again a conditional distribution.

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So earlier, how did we model this? We just modeled it as the following we said that the whole context of y1 to yt minus 1 is just contained in that blue vector, which is st right. So, remove this variable and replace it by a vector does that make sense? Ok.

Now you have the image also. So, how are you going to model this? So, what are you going to write on the right hand side? Ok let me give you a hint, we all agreed that this is the quantity that, we are interested in right. We also agreed that the following is fine replacing y1 to t - 1 by st is fine. Now what about the image? What do you mean by objection in the image? You will supply the words, which are there the object names that man fine that is fair enough well, if want to make it more abstract more neural. So, what you are saying is that whatever information is contained in the image should be passed here.

Whatever information is contained in the image should be passed here, how do you what's the way that you have learnt of computing the information in the image a dash neural network?

Students: (Refer Time: 14:15).

A?

Students: Convolutional neural network.

Feedforward neural network?

Students: Convolutional neural network.

Convolutional neural network ok. So, but what from a convolutional neural network? How many representations that is a convolutional neural network learn? How many does v g g network learn? v g g 16, the last layer is a softmax layer 15 right. So, which one will you give now? One before the last one, that is called the?

Students: (Refer Time: 14:36).

Dash layer, dash dash layer, fully dash layer.

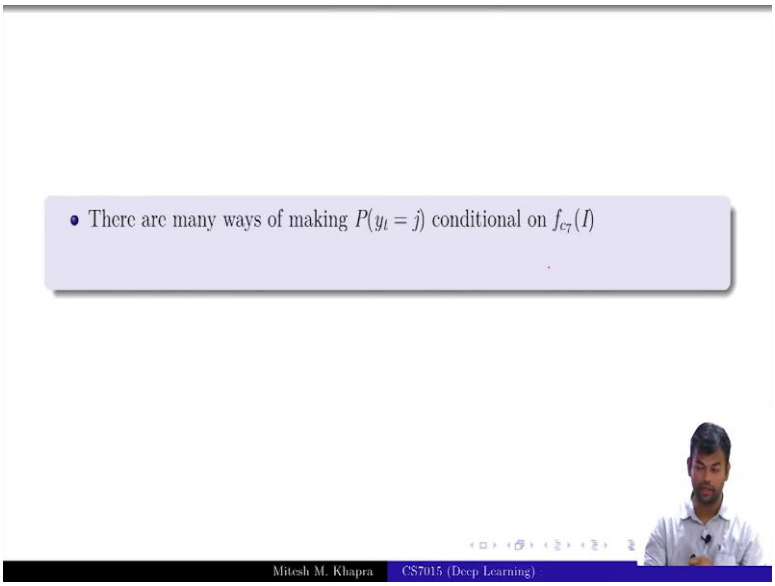
Students: Fully connected layer.

Fully connected layer ok, at least your language moral works fine ok. So, that is the fully connected layer. Remember that all the layers in the convolutional neural network, learn an abstract representation of the image and as she was trying to say, that this abstract representation contains or at least you believe it contains all the information that is there in the original image, just as st contains all the information that was there in the sequence y1 to t minus 1, this abstract representation that we will get from a CNN, contains all the representation all the information, that is there in the image we all believe that ok.

And we also believe that any of these representation is fine. In practice, the convention is to use the fully connected layer that is called as f c 7, the seventh fully connected layer right and it is 7, because you also start the numbering from the convolution layer 1, 2, 3, 4, 5 and then the seventh layer ok. So, that is what you will take ok. So, does this make sense and it is a very simple extension from what we were doing earlier, this is what I have circles is what we were doing earlier right, where we only had st. Now, I am saying is just as you believe that st and codes all the information in the previous sequence.

I am just asking you to stretch that a bit more and say that f c 7 of the image contains all the information that was there in the image is it fine?

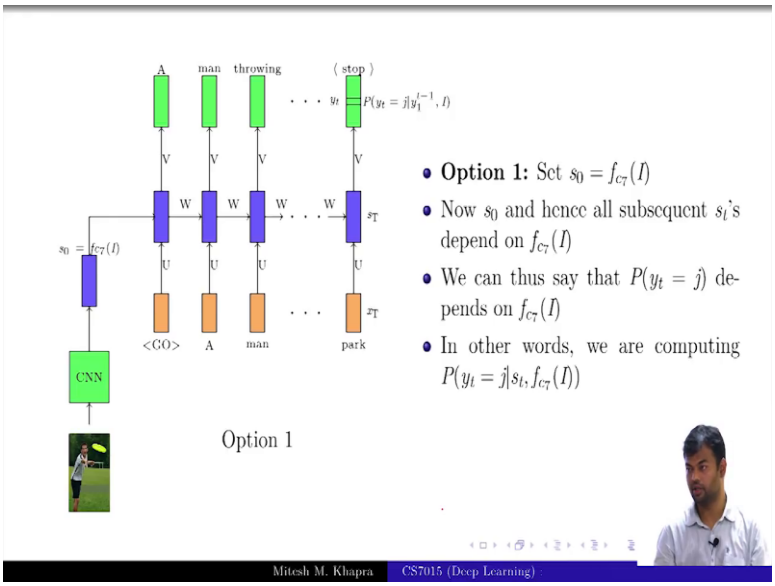
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Ok, but still there are some issues and there are other ways of making this condition on f c 7, in particular what you could have done as she was trying to suggest initial is that maybe you have a vocabulary of all the objects that are possible in your image right. So, maybe in your image there is man, woman there is flying desk, frisbee or there is dog, cat and all these things right.

So, you do an object detection first, get out all the object which are there and then make the distribution conditional on these objects right. So, you can say that I will allow for a 10 words to describe the image. So, there word 1 is equal to man, because I have detected the object man in the image word 2 is equal to Frisbee, because I have detected the object Frisbee, in the image that is all that is another way of doing it ok. So, I just want to make it clear, that there are different ways of making the conditional distribution, conditional on the image itself, we are choosing to make it conditional on f c 7 of i right that is the neural way of doing it ok.

(Refer Slide Time: 16:59)



So let us see, 2 such options, the first thing that we could do is we could set s_0 to f_{c7} of the image, what is s 0? The first thing that was passed to the language model ok. So remember, we had this go symbol and we had this s 0, which was mysterious we did not know how it comes, but now we know it that s 0 could just be the image that is what my starting point is. So, take this image and now start generating the representation generating a description does that make sense? Ok. So, this is what the network looks like.

So, what do you saying is that these things are of dimension d, the CNS output was say of dimension 4096. So, this has to be converted to size d right; that means, what will you, how will you do that? We have a 4096 dimensional vector and you want to convert it to a d dimensional vector w belonging to?

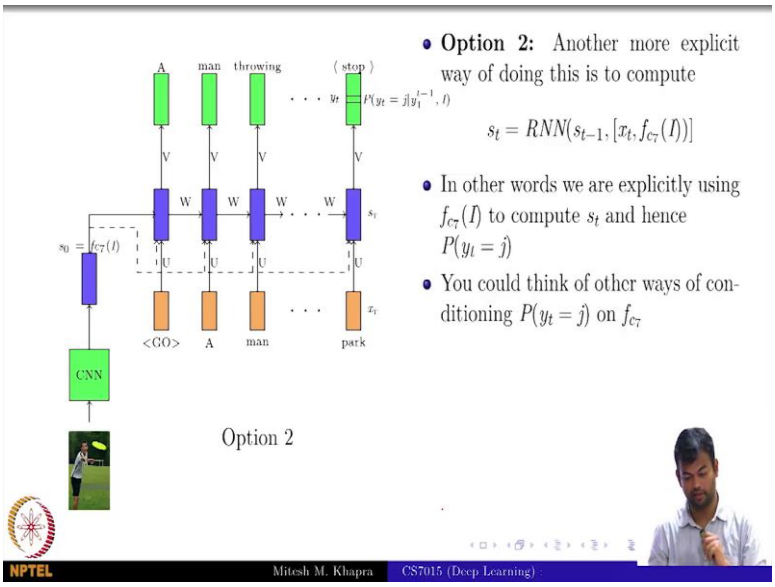
Students: (Refer Time: 17:51).

4096 was d. Fine in general, any 2 vectors if you want to make them compatible, this is what you will do you will project. So, that they are of the same dimensions. x 0 will be the go symbol. Go is the special word in your vocabulary, which says star generating the sentence right. So, whatever vocabularies you will add 2 special words right one is go and other is stop. So, whenever you generate stop, you stop generating after that fine. What is the other way of. So, here now what happens is. So, this is what is happening technically and that is why that is what I wanted you to understand this now s 1 depends

on s_0 ok. What we are interested in is the following that yt should be conditional on y1 to t minus 1 comma image ok.

They have make sure that i is s 0 and this quantity is st, you have to find now since the first time step depended on the image, all subsequent time steps will depend on the image, is that ok? What is the other way of doing this? What now in this looks slightly inefficient, what's the other option that you could have used? Just feed the image at every time right. So, that is the one constant thing that this is the image, now whatever you have generated so far, considered that, but in the addition to that also consider the image.

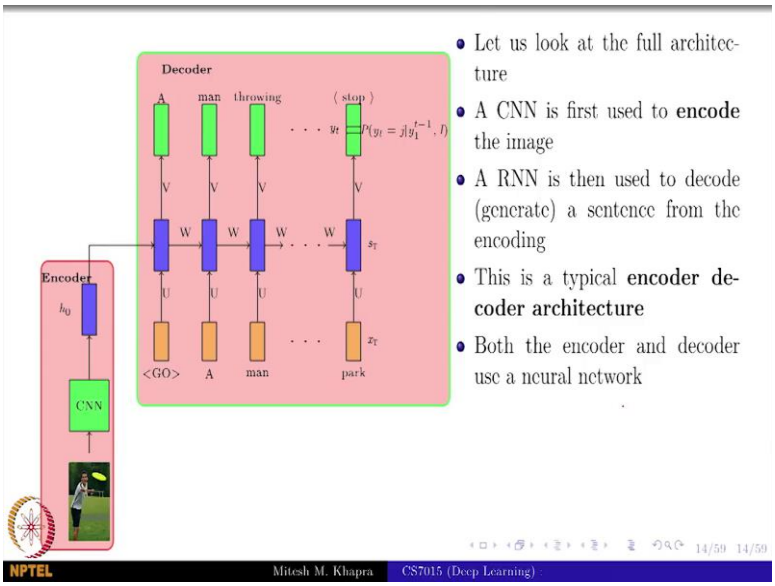
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So, what would the diagram look like just passing, the input to every stage of the decoder ok.

I have already started using terminology, which have not introduced, but I will just introduce it shortly.

(Refer Slide Time: 19:21)



So, let us look at what the full architecture looks like, there is something known as the encoder, which takes your input encodes it and gives you a representation right. Then you have something known as a decoder because, given this input you want to decode what the output is right. So, remember general terminology would be whatever input is given to you, you want to encode it and whatever is the output that needs to be decoded right it is you could think of it that, this is the image. Now, I am trying to decode the description for there, is that fine? Ok and then you have an RNN, which is used to decode the sentence from this input.

So, such architectures are known as encoder decoder architecture and these are become extremely popular and we will see why they are so popular and why they have led to the popularity of deep learning in general ok. So, everyone understands this diagram, anyone who does not see a problem with this diagram? There is actually, no problem, but I want you, I want you to see beyond the diagram and to look at the equations, what do you mean by that? What do I have here as the input? What is my x? So, this this looks fine, I have taken 1 box and connected it to another box and everything is fine right, but that is not what I am interested in.

What am I interested in? Can you write the input as a the output as the function of the input in this case? Is it possible to do that? So, that is what we need to make sure that we are able to do right. So, we look at various applications suggest lepted criptical here, but

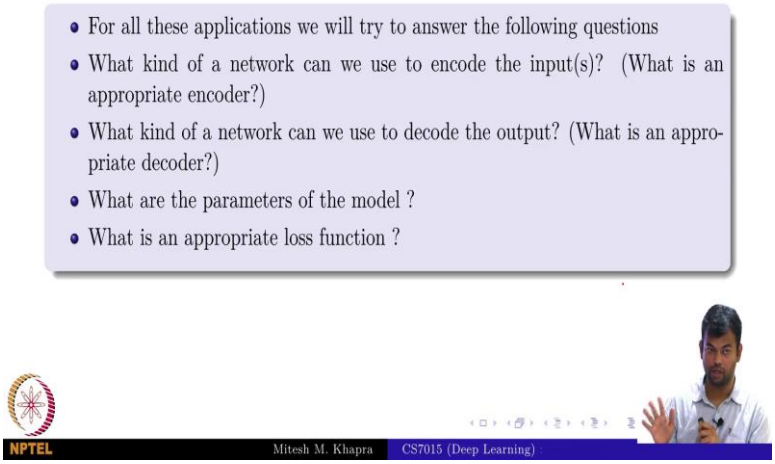
I am going to come back to it. So, I just the emphasis that look beyond the diagram, the diagram looks very nice. I hope it does thanks to the TA's, but it does. And, but we need to understand, what is the what is the set of equations being conveyed through this diagram right? What is the function that we are trying to learn? We are going to write y as a function of x, are we able to write that function? Because, now we are suddenly thrown in a convolution neural network at some place, we have an recurrent neural network then, we have the feed forward layer at the output, which is the green vectors. So, does all this combined together right.

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Lecture – 113
Applications of Encoder Decoder models

So, we are going to see a lot of Applications of the Encoder Decoder models.

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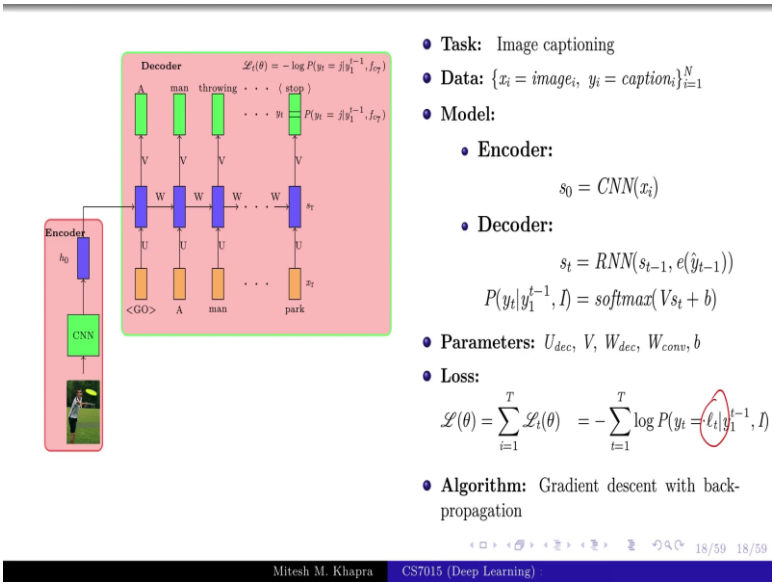


And, for all these applications we are trying to answer the following questions. What kind of network can we used to encode the input? In the previous application what do we use? CNN. What kind of network can be used to decode the output? What did we use?

Student: RNN.

RNN. What are the parameters of the model? We will see that and what is an appropriate loss function, right.

(Refer Slide Time: 00:37)



So, let us again go back to this task which was image captioning. What is the input what is the training data given to you? What is x? What is y? x is the image, y is the description, right. So, this is what is given to you given N such training pairs where x i is the image and y i is the description and y i in itself is a sequence, right. So, you have y i 1 to y i capital T, everyone gets the input and output?

Now, what is the next thing? Model. Can you write down the model equations? I want an equation which starts from x and goes all the way up to y and since we have several time steps I want an equation for y t, this is generic for every time step right say can you write that equation and feel free to use shortcuts. So, you do not need to write the entire RNN equation just say RNN of something. Dont even try to write the VGG 16 CNN equations just say CNN ok so, we will go ahead.

The first thing that I am going to do is, I am going to write the equation for the encoder. So, the encoder gives me CNN of x i whatever is the input given to me, x i is the ith training image given to me. So, I will just pass it through CNN I will get a representation for that and I am just being cryptic here it could be the f_{c7} representation or the con 5 representation or the max fool 5 representation or whatever you want, right and it is going to denote all of this as CNN of x i, run this CNN take whichever representation you want to take.

Now, what is the decoder going to be? Decoder is the following RNN. Remember the equation of RNN was s_{t-1} comma x t, what is the input of the tth time step? Whatever we are predicted at the previous time step, just the embedding of that. So, e means embedding, if you want take one odd embedding if you want take word to vec embedding, is that fine? Ok?

And then what is the output? It is the soft max function of the following. How many of you get this now please raise your hands, how many of you can say that y can be written as a function of x, is that pretty straight forward ok. So, you have an encoder, you have a decoder and remember that this final y is a composite function of the original input x ok. Just that you are doing too many computations along the way, but there is a path which exists ok.

What is the loss function, everyone at this point should be able to say it?

Student: sum of cross entropies.

Sum of cross entropies. They just wait for me to say two more sentences. What are the parameters U V W B C?

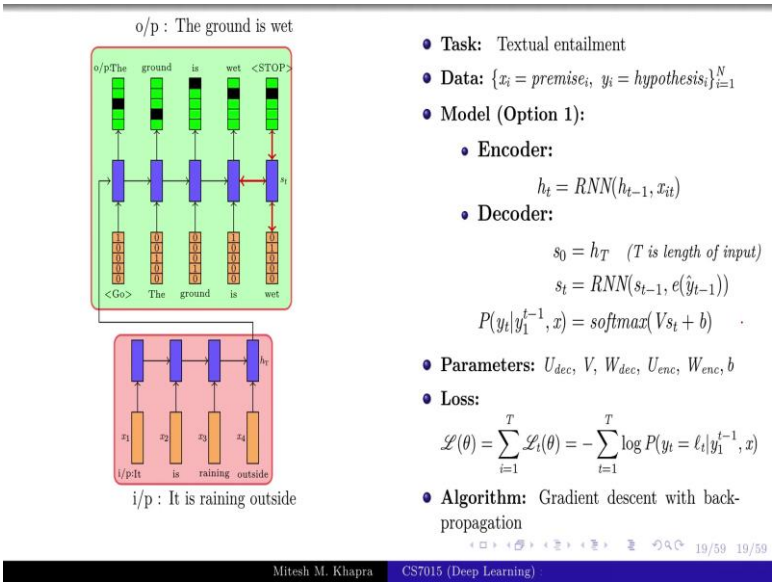
Student: (Refer Time: 03:12).

A B C D E F [laugher] everything, right. What is that? All the parameters of the?

Student: (Refer Time: 03:17).

Convolutional neural network; that means, all the filters that you have, all the parameters of the RNN which is W, U and the parameters of the output layer which is V, right, all of these. Is that fine, I am I may have missed some biases but ok. The objective function as you said is a sum of cross entropies, where l t is the true character at time step true word at time step t. And what is the algorithm that you are going use back propagation through time and with back propagate all the way through the CNN also which is an end to end thing. In practice of course, you do not do that yes you could just said both to be the same. Do you get that question? Is that ok? Ok.

(Refer Slide Time: 04:00)



Now, let us look at another task we look at the task of textual entailment. What textual entailment does is that I give you a input or premise the premises that it is raining outside and you need to tell me a hypothesis. The hypothesis that the ground it is wet, ok. With basically means that it is raining outside implies that the ground is wet ok. Now, what is the encoder decoder architecture that you will use for this problem? What is the input here?

Student: A sequence.

A sequence. What is the output? Sequence. It is all the hint that I am going to give you. So, what will you do what is the encoder equation going to be?

Student: RNN.

RNN. What is the decoder equation going to be?

Student: RNN.

RNN. How will it become end to end by setting what to what?

Student: (Refer Time: 04:49).

S0 of the decoder to, to what of the encoder?

Student: (Refer Time: 04:49).

Last time step of the encoder. How many if you get that? Really we are on the same page first time in I do not know how many lectures by finally, it happened. So, here is what training data is, right it is a collection of premises and hypothesis and you have N of these. There are two options for the model the first option is that you encode the input using an RNN feel free to replace and by an LSTM if you want, then you have the decoder where and you set the zero time step to whatever you got from the encoder.

Then every time step you computed using the RNN where remember the input at every time step is whatever you predicted at the previous time step and then the output is just the soft max function, is that fine? And what is the loss function going to be, loss function?

Student: (Refer Time: 05:44).

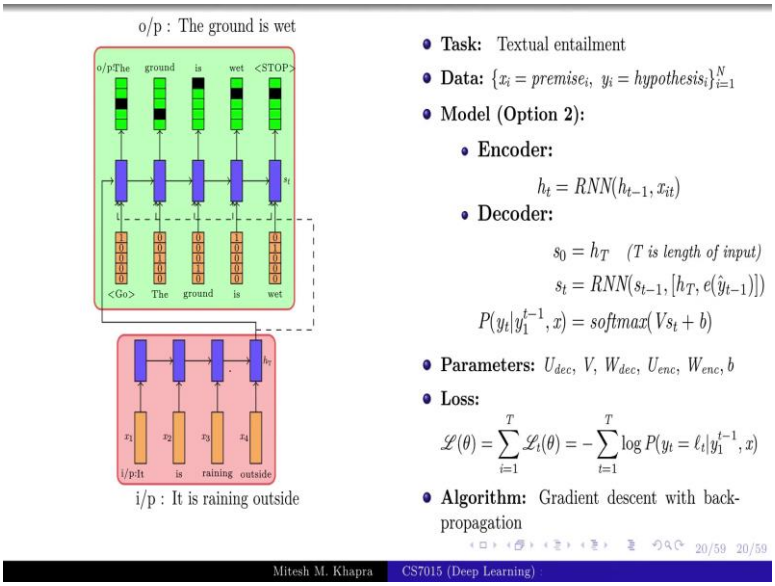
Sum of cross entropies. Training algorithm?

Student: Back propagation.

Back propagation through time and really it is through time right all the way back ok so, we will see that. Let me see if I had any other question ha ok I will ask it parameters I am not going to bother about ok. Now, this was option 1 I have just clearly written, what is option 2, what is the set of equations look like for option 2? Which of these equations will change and how? Remember option 2 was maybe pass the input at every time step, which equation will change?

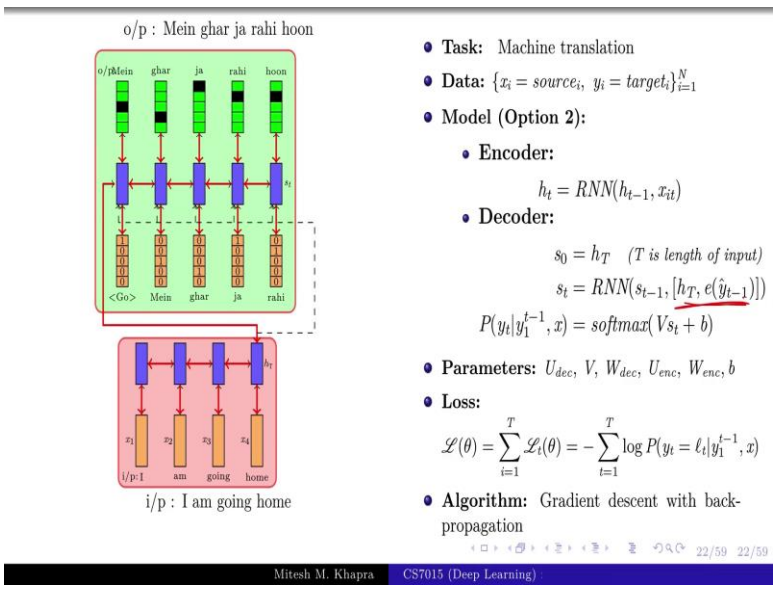
St, what will it become? But s t can take only I mean RNN take only two inputs, right. S t minus when you need to gave embedding you need to gave so, how will you fit in the third input, this animation has it is own mind. So, this is how back propagation will happen, right so, let us it is finish that. So, will actually back propagate all the way back through time, fine really all the way back through time.

(Refer Slide Time: 06:40)



And same task textually entailment I want model 2 option 2. So, this is what will happen. We will just concatenate h capital T which is this guy along with the input at every time step, right. How many if you get this the RNN is still taking just two inputs? One is the previous state the other is the concatenation of the current input as well as input that we got from the encoder. Everyone get this? Ok. So, this is model 2 I am going forward I am not going to do both model 1 and model 2 it is model 2 is just a very simple variation of model 1, a parameters loss function training algorithm everything remains the same ok.

(Refer Slide Time: 07:20)



Let us look at machine translation. What is the input? An English sentence. What is the output? A Hindi sentence. What is the encoder going to be?

Student: RNN.

RNN. What is the decoder going to be?

Student: RNN6RNN. What is the loss function going to be?

Student: (Refer Time: 07:37).

Soft max. Who said soft max?

Student: (Refer Time: 07:42).

What is the loss function going to be?

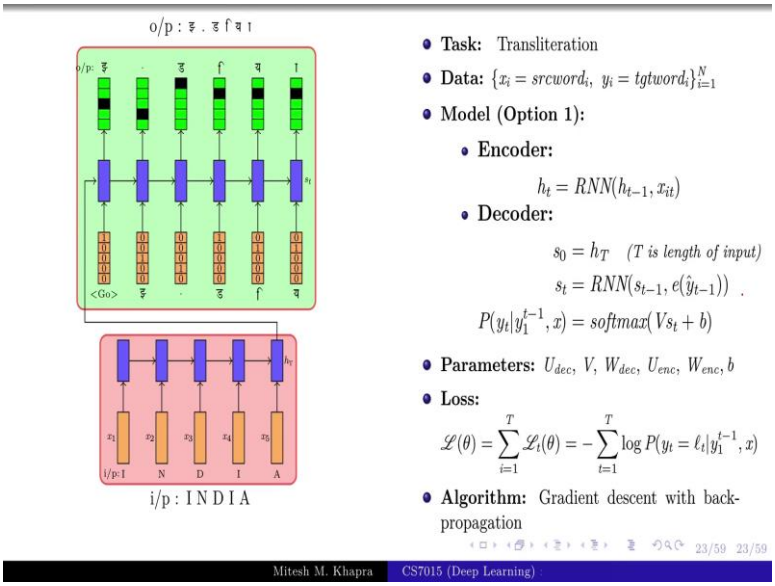
Student: sum of.

Sum of cross entropies, training algorithm all the way through time, right ok. So, let us can you draw can you write the equations? Just copy it from the previous slide, right actually copy it from the previous slide, right. If you have the RNN you have the RNN as a decoder, again in option when you will set s 0 to h t you have the loss function the parameters and your training algorithm, ok. And for option 2 it is back propagation will, fine and for option 2 what will happen? Option 2 what will happen?

Student: (Refer Time: 08:17).

This will change, right. So, just focus on that we just passed in the last time say belong with that.

(Refer Slide Time: 08:27)



Now, transliteration, what is transliteration? What is transliteration, if you do not know it at least see it from the example what is it?

Student: (Refer Time: 08:34).

Writing the same word in another language, right. So, this is typically done for named entities very when you are; when you are translating from one language to another you do not often for Thomas you do not come up with an Indian translation right you just say Thomas in Devanagari, right you just right Thomas in the Devanagari, right. So, for names you typically just do a transliteration; that means, from the English script you just write it in the native language script ok. What is the input? One word the input is the word right. What is it, a sequence of?

Student: Characters.

Characters. What is the output?

Student: Sequence of characters.

A sequence of characters. What will you use for the input?

Student: RNN.

RNN. What for the output?

Student: RNN.

RNN. So, I will becoming too easy, right. Can you write the equations for this? Yes, you will copy it from the previous slide, yes ok. Everything remains the same, right. So, you see why this framework has become so powerful, you do not see it still maybe let us look at something else.

(Refer Slide Time: 09:33)

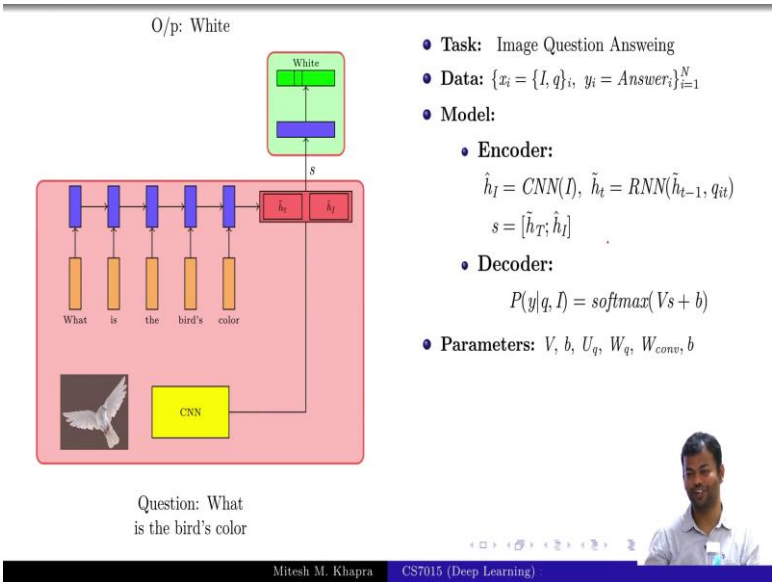


Image question answering; tell me what is the data here? What is the input? Image and?

Student: Question.

Question and what is the answer, what is the output? Answer. So, for simplicity we are going to assume that the answer here is a finite vocabulary. We are not generating descriptive answers, we are not being overly dramatic let us going to say one word what is the colour? white. We are not going to write I think the colour of the image is white now ok, just white. So, all these outputs are going to be single words and we have v possibilities and we are going to predict one of those v possibilities ok.

Now, give me a model for this, now, things are getting slightly complicated you have one image as the input one sequence as an input and a dash as the output. Oh god, now think why would you generate the sequence of characters as the answer. I said that the answer is going to be come from a finite vocabulary; that means, you need a?

Student: probability distribution.

A distribution, probability distribution is here enough said. Now, tell me what is the model? A model should connect the input to the output. You have two inputs here. I see some people doing this [laugher] I do not know what that means, but let us just do it let us make a train. Simple formula simple recipe and whatever input you are given just encode it. Depending on the type of input you know what is the encoding is going to be. For images what is encoding, sequences?

Student: (Refer Time: 10:55).

Now what do you do with these two separate things?

Student: (Refer Time: 10:58).

Concatenate them. Ok and then?

Student: (Refer Time: 11:01).

After that?

Student: (Refer Time: 11:02).

Can you think of all the equations can imagine all the equations along the way?

Student: Yes.

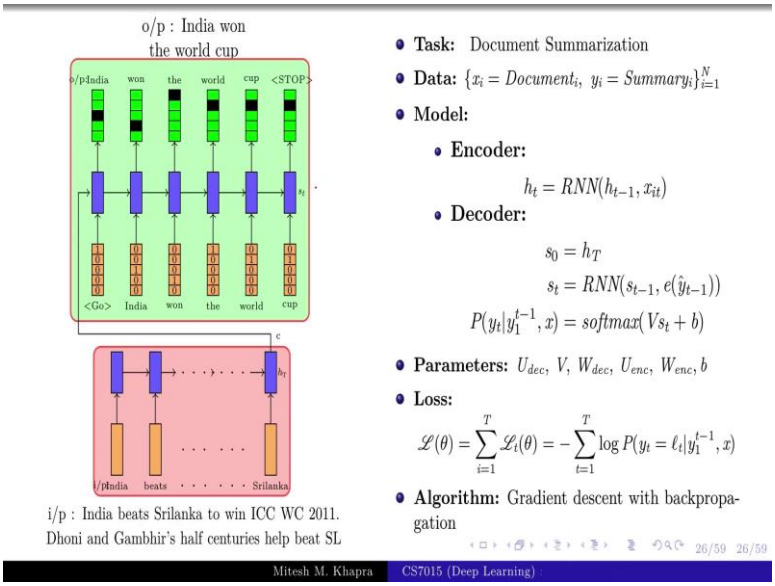
Of course, yes [laugher] right I mean imagination [laugher] you can always do that [Laughter]. Now, just think about it. Can you write the output as a function of the input where the input is actually a pair now, it is image comma question what is the model going to look like let us see. So, model will first have an encoder for the image let us call that as $\hat{h}_i$, it is going to have an encoding of the question let us call it as it as $\tilde{h}$. I am going to concatenate these two as someone rightly gestured and then what am I going to do after that pass it through a?

Student: Feed forward network.

Feed forward network and predict a probability distribution. What are the parameters of this network? Parameters of the feed forward network, the parameters of the recurrent

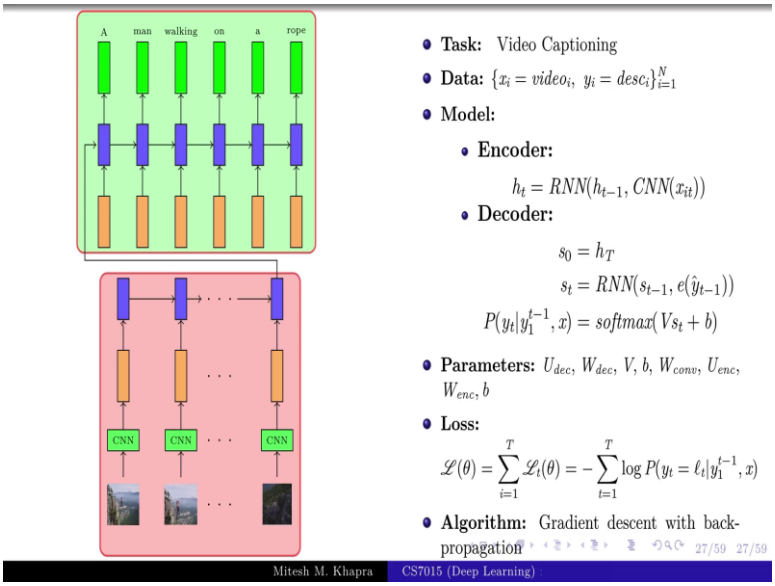
neural networks and the parameters of the CNN, right. So, everything that we have done so fine because ok. How do you train it? Back propagate through time and space, ok, also go back to the image also, fine? Is that ok? Ok.

(Refer Slide Time: 12:09)



Document summarisation; what is the input? Sequence. What is the output? RNN, RNN everywhere, fine. I will not even bother to ask you

(Refer Slide Time: 12:18)



Video captioning; sequence of images. I want to hear the choice of phrasing that you use. I just want to hear that I love hearing that every time.

Student: (Refer Time: 12:30).

RNN of CNNs whatever that means, what RNN of CNN. Every time I do this everyone says RNN of CNN [laugher] I do not know what that means, but it is the right answer. What does it mean? What is the video? It is a sequence. What is the sequence of?

Student: Images.

Images. So, what will you do, encode every image and then pass it through a?

Student: RNN.

RNN can you imagine the equations, ok, let us see. Ok and in this case what is the output again? The sequence. So, what is the decoder going to be?

Student: RNN.

So here is the model. So, first what you do is for every time step you compute the CNN encoding of the frame, then you pass it through an RNN to get the final time step t and then you feed it to a decoder and generate one word at a time, is that fine? So, this thing apparently is called RNN of CNNs ok. And so, that is and loss function would again be the same sum of cross entropies and back propagation through time and space, ok, good. Please do not quote me on this thing also this is getting a recorded but ok.

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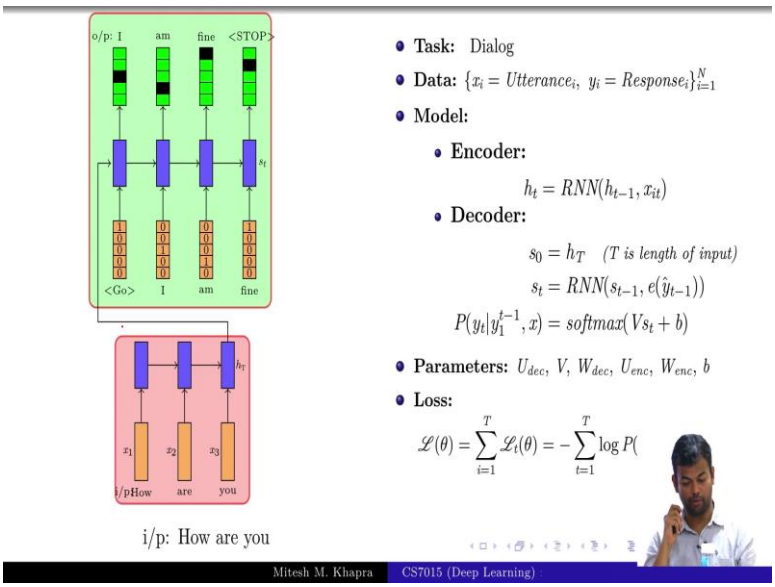


The next one video classification, what is the decoder? Decoder is probability distribution, okay, what is the decoder?

Student: feed forward neural network.

Feed forward neural network.

(Refer Slide Time: 13:54)



Ok. This one, dialogue how are you I am not fine, ok, input?

Student: RNN.

Output?

Student: RNN.

Right. So, you see this why this has become so, popular we took a wide range of problems different modalities right we took images, we took videos, we took sequences, a combination of these. Right? Image question answering has a combination of images and sentences and the output you have a probability distribution. All of this could be model by this unified end to end network, all of the components are neural network based components whether it is a convolutional network or a feed forward network or a recurrent neural network, right.

Now, let me stretch this right what if you have video question answering. What is the input going to be sequence of images and sequence of?

Student: Words.

Words. The output is?

Student: (Refer Time: 14:46).

No. Just a word right we will pick from a fixed vocabulary. How are you going to model the input?

Student: (Refer Time: 14:53).

RNN for the question, RNN of CNN for the video, then?

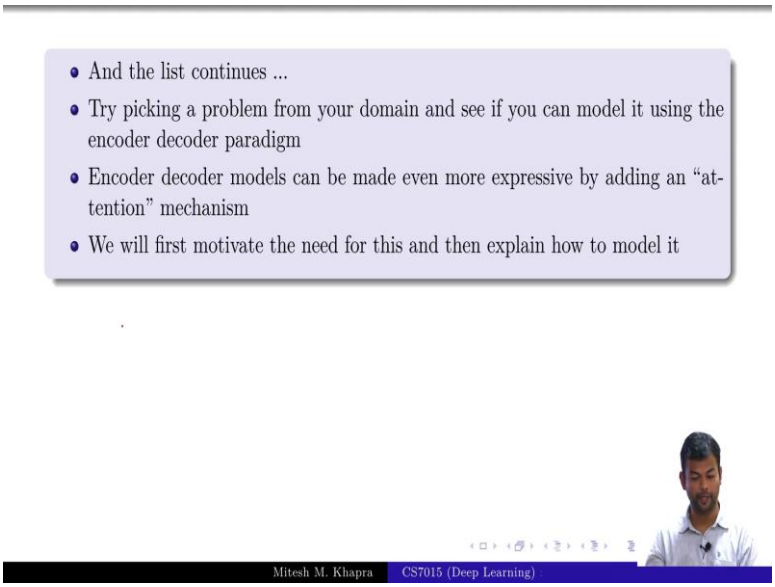
Student: Concatenate.

Concatenate and?

Student: Feed.

Feed it to your feed forward network right. So, all of these become to what extent that work is a separate question, but all it was not even possible to model all of this as an end to end network, right. But, now I just because possible to model it as an end to end network with all the components being neural components, right. And, this is this story still not complete everything is not as easy as it looks they still and very crucial component that we have missing in the architectures that we have seen so far and we will talk about that soon ok.

(Refer Slide Time: 15:33)



Now, let us just continue that I challenge you to do this, pick up any problem there are student from relevant different departments, pickup any problem do not say that I want to design some gear for certain aeroplane and all that and I want to use neural network to do so. No. Something which involves machine learning, right not problem is does not involve machine learning and see if you can model it using the encoder decoder frame work. just try to do this.

Take problem from biotech right for example, they given a sequence of genes and you want to predict whether this person is susceptible to a certain disease, what will you do? Conduct a blood test ok. Do not do not [laugher] do not go and do neural networks for that, but if you had to do that this is what you will do it will take a sequence you will treat the sequence of sequence at the given DNA as a sequence of genes and then you will try to predict something as the output try to predict a probability distribution over disease it a possible, right.

So, you can think of many applications from many domains and all of that you could problems involving machine learning with potentially model than using the neural encoder decoder architecture. But, there is a very important part missing from this whole story which is attention which is a very important idea and we will spend some time on that in the remainder of the lectures. So, we will first motivate why do we need attention

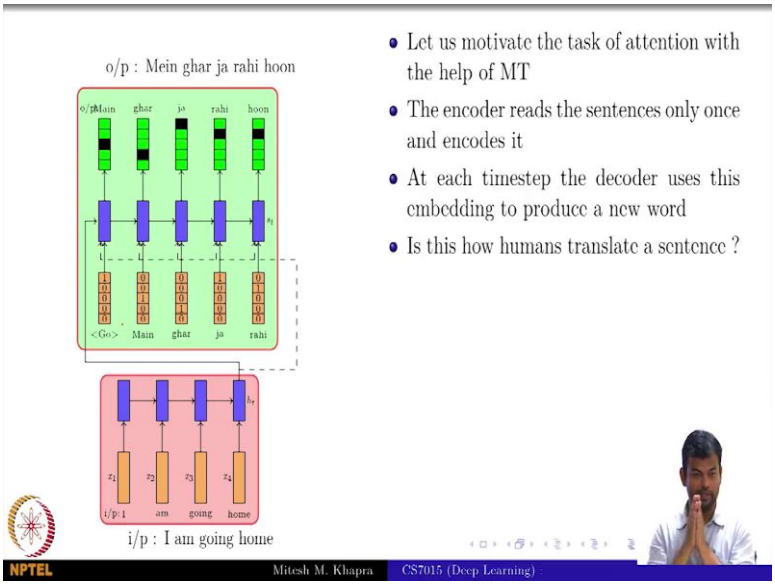
and from there we will see that how do you make how do you integrate attention with all these encoder decoder architectures that you have seen so far.

Deep Learning
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Lecture – 114
Attention Mechanism

So, let us go on to the next module which is Attention Mechanism.

(Refer Slide Time: 00:14)



So, let us motivate not the task of attention, let us motivate attention mechanisms with the help of machine translation, ok. So, what is happening in the models that we have seen so far? The current model that we saw for machine translation; by the way all the models that I have shown you so far are wrong or rather incomplete we will complete all of them, right and that is where attention fits in, ok, that was for the camera.

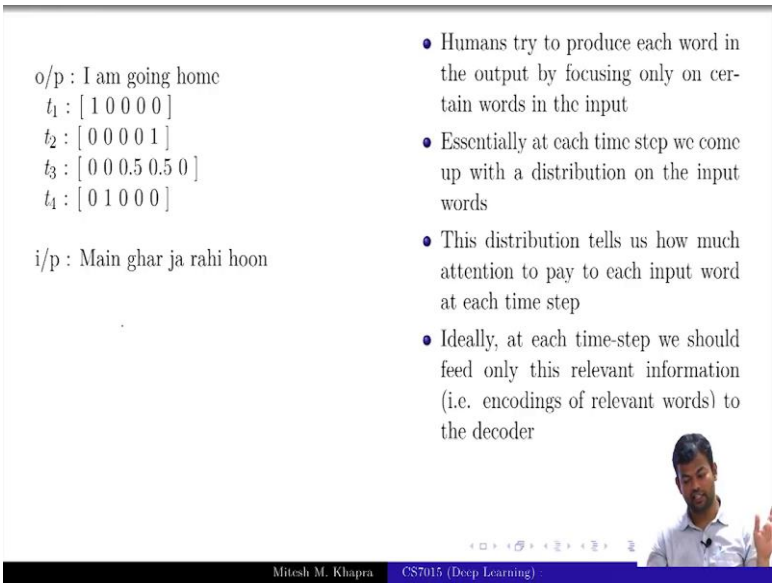
A encoder reads the sentence and its computes the encoding once, right we read the entire sentence and be encoded it. And then we have these two options either the pass the encoding at the 0 time step or pass this encoding at every time step. Is this how humans translate a sentence? What is the human analogy for this? You have read the sentence once done and now we are going to remember this entire thing throughout and then translate. Imagine if you doing this for sentences which have 25 words which is a typical Wikipedia sentence. What is wrong with this? We have read the input ones and we have

encoded it, what is likely to happen you will forget something you are going to lose information. Not just that is the entire sentence important that every time step?

Student: (Refer Time: 01:28).

Only certain words are important. You see this conceptually something wrong that we are doing here, is is saying, ok. I have encoded the sentence and then start decoding from there, ok. Thats the conceptual error that we are making. So, let us see how humans actually try translate it, right.

(Refer Slide Time: 01:39)



So, when producing one word in the output suppose my input is the Hindi sentence and I have the output sentence. When I am trying to produce the first word I actually compute this probability distribution which tells me which of the input words that I need to focus on. At this point it is ok if I dont know what is the translation for ghar or ja or raha or hoon, as long as I know the translation for mai I am done because that is the word which I first need to produce there, right.

So, I am going to say that at this point I only need to pay attention with the first word in the input and I can ignore everything else. What about the second time step? I just need to focus on the last word. What about the third time step? Is it always going to be that I only need to focus on one word at a time?

Student: No.

No. What about the third time step?

Student: [FL].

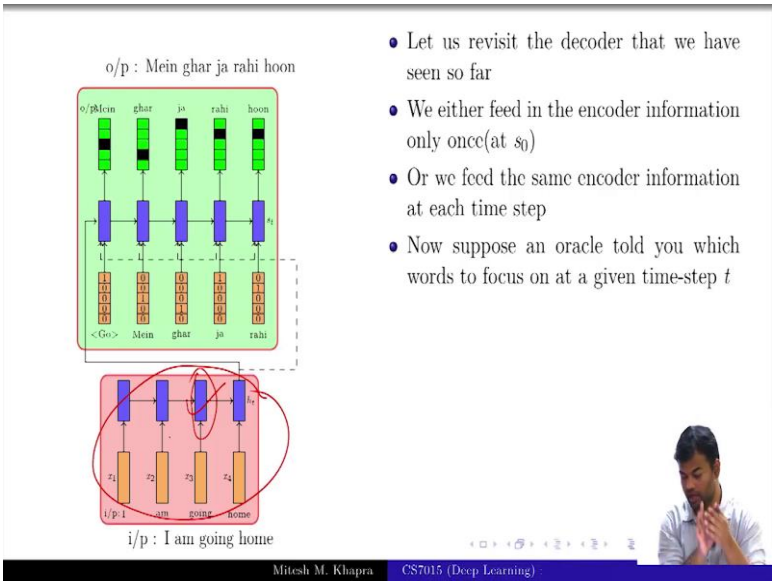
I am sorry I am assuming everyone understands Hindi, but I think that is this is small sentence I can assume that. What will you focus on?

Student: Ja rahi.

Ja and rahi, right. You want to focus on both these things and not an anything else and what about the next one? Hoon, right; so, just on ghar and not an anything else. Is this what the model encoder decoder model is doing? What is it doing actually? The every time step is focusing on the entire sentence because that is the encoding that your feeding to every time step, that is the problem that we need to correct. We need to learn to pay attention to certain important parts of the sentence. Is the setup clear to all of you? Is the motivation fine? Not your motivation layers; is the motivation for this fine or not, ok.

The distribution actually tells us how much attention to pay to each input word at each time step, and ideally at each time step we should face pay attention to only certain words in the input.

(Refer Slide Time: 03:22)



So, let us revisit the decoder that we have seen so far. This is what the decoder looks like. In fact, I also have the encoder there. Now suppose, sorry. So, currently what we are

doing is we are either feeding s 0 at the I mean we have either feeding the input embedding or the encoder embedding at time step 0 or at every time step; the suppose there was an oracle which told us exactly which are the words important at time step t, right. So, in our example at time step 3 suppose it told us that the word going is important, actually we need to flip the input and output here also, right. But you can still understand, right.

So, I am saying at time step 3 certain words are important and suppose a oracle actually told us that these are the words which are important. What would you do? Assume that you have already run the encoder, what will you do now? And say someone told you that only this word is important word why weighted I am just saying binary weigths, right only this word is important. What would you do ideally?

Student: (Refer Time: 04:24).

Just feed this blue vector to the decoder and do not feed everything else, does not make sense. Suppose I told you that two words were important send those two words but how concatenate, but now at certain time steps 4 words will be important, and you cannot concatenate 4 words, right because then the dimensions will change. So, what do you do?

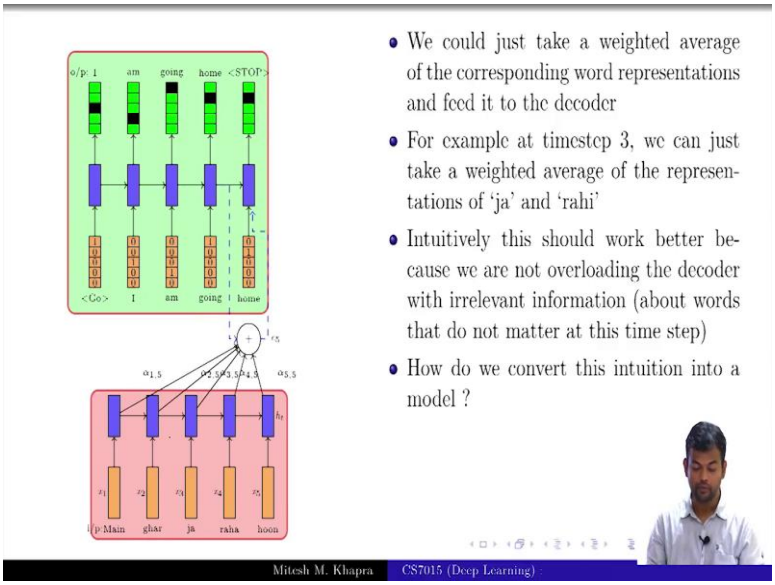
Student: (Refer Time: 04:47).

A weighted.

Student: (Refer Time: 04:49).

Weighted some of the important inputs is that make sense. At time stamp 3 we saw that ja was 0.5 important and rahi was 0.5 important, just a weighted combination of those two blue vectors and feed that to the decoder. So, you are not changing the dimensions at each time stamp because the blue vectors have the same dimension, I am just taking a weighted combination of those I am going to give you the same dimension; does that make sense, ok.

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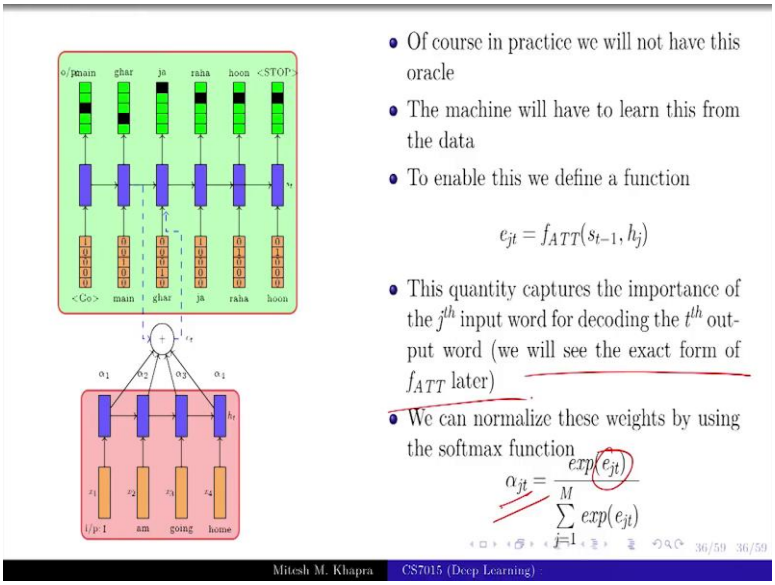


So, in fact what I am saying is that, I could just take a weighted combination of all the blue vectors that I have at the encoder and the weights of this weighted combination, right now I am assuming that someone oracle has given me is that, ok. If I had his weights does this makes more sense then having the vanilla encoder decoder model, everyone agrees with that, ok.

Now, the question of course, us who is going to give us these weights, we will come back to that later, but at least given the weights this make sense. So, at every time step they just going to focus on the words which are actually important, just take a weighted combination of those words and we will just feed that to the decoder. And intuitively this should work better because unlike before where we were overloading the decoder with the entire sentence remember 25 words, 30 words, entire sentence was being passed to the decoder now you are just overloading it with the amount of information that it actually needs to produce that particular word. Hence, intuitively this should work better, right, ok.

Now, how do you convert this intuition into a model?

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In practice of course, there is no angel who is going to coming give as these weights is no oracle. The machine will have to learn this from the data. Whenever you need to learn something you need to.

Student: (Refer Time: 06:34).

Introduce parameters. So, I am going to now introduce a parametric form for the from the figure which thing for those of we cannot see these are alpha 1, alpha 2 and so on. So, now, from the figure we are going to introduce a parametric form for.

Student: Alphas.

For the alphas, ok. So, I am going to introduce I will come to alpha. But and what you think this weight should depend on. What I am trying to say is that at the tth time step of the decoder, so this is e_{jt} at the tth time step of the decoder I want to find out how important is the jth word in the input. That is exactly what I am interested in at every time step of the decoder of all the input words I want to see which of them is the most important, right. So, this is the quantity that I am interested is, how important is the jth input word at the tth time step. This should depend on what? What should it be a function of?

For one it should depend on what that word is, right the other is should depend on what has happened in the decoder so far. What is the decoder produce? So, what is the input

and what is the decoder state at so far, right. So, as the decoder has already decoded the word ghar or home, it does not need to look back at home, right. That is why need to know what is the state of the decoder. What captures the state of the decoder at time step t? H_t. And what is the state of all the words that we have? It is captured by what? The h_j's, right, this is h 1, h 2, h 3, h 4. Does that make sense? How many of you have fine at this point? Please raise your hands high above, ok. How many of you have questions, please ask specific questions if you have a question.

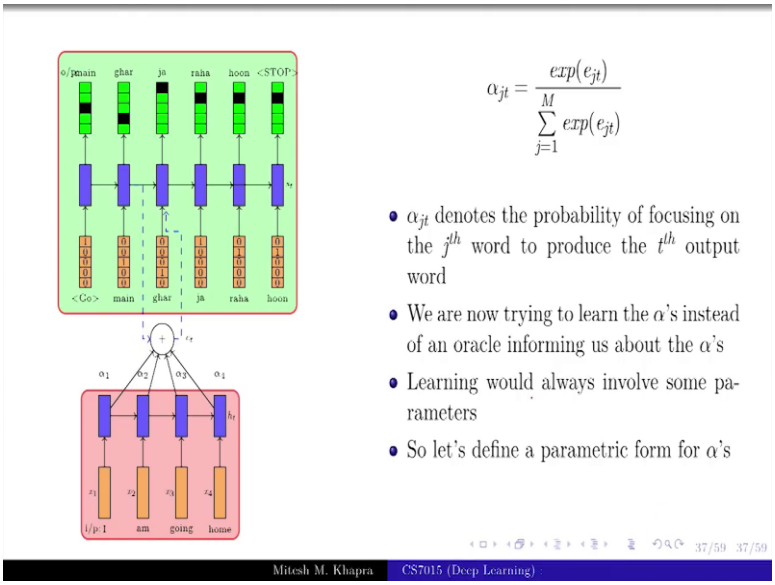
All I am saying is a couple of things one is at every time step instead of the oracle giving me these weights I want to learn these weights. Whenever I want to learn something I have to introduce a parametric form and then I learn those parameters, ok. Now, what is the quantity that I am interested in? I am interested in this for all the input words, for the jth input word I am interested in knowing how important it is for the tth time step. There are several ways I could write this function, I am saying that the two things that are important is one what is this j^{th} word which is captured by h_j right and what is the state of the decoder up to this time step which is captured by s_{t-1}.

You could think of various other equations. At this point I am fine if you by the intuition that this quantity should indeed depend on these two terms it should depend on what has happened in the decoder so far and what is my current word actually look like. How many of your fine with that please raise your hands up and high? Ok. Now also the other thing that I want is that across all the input words this should actually sum to 1, right. I just want a weighted combination I do not want arbitrary weights it just like taking a probability distribution over what which word is important by how much. So, if I have this e_{jt} how will I convert it to a probability distribution?

Student: Softmax.

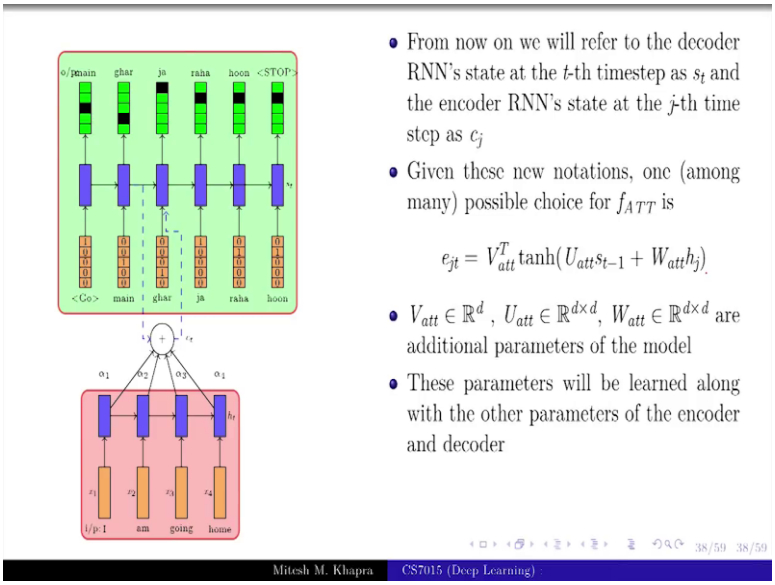
Softmax. So, I will just compute the alpha j's as a softmax of e_{jt}, e $_j$'s is that fine; did not get this, ok. Now, we have still not seen what the exact form of attention is, of the f attention function is.

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So, this is what the equation for the α_{jt} is that we had an alpha j actually denotes the probability of focusing on the jth word at the tth time step, ok. Now, we are now trying to learn these alphas instead of an oracle telling us what these alphas are. So, learning is always going to involve some parameters. So, let us define a parametric form for alphas and just a couple of notations.

(Refer Slide Time: 10:25)



So, from now on we would not change this. We are going to refer to the decoder state as s_t and the encoder state as s_{hj}, ok. So, these blue vectors are s s and these blue vectors are h_s, ok.

Given these new notations one among many possible choices for f attention is the following. I wanted it to depend on the current decoder state I am making a dependent on the current decoder state but I am also adding a parameter in front of it, right. I also wanted to make a dependent on h j I am making a dependent on that I am also adding a parameter here.

Why do I need this parameter? What is the dimension of this? Let us assume this is also what is the dimension of this. Remember after multiplying with u attention and after multiplying with w attention the two vectors should be addable, is that fine. Something cross d what about this same thing cross d, good, ok. So, let us call that same thing as d 1. What is this output then? The tanh output is vector scalar matrix vector of size.

Student: (Refer Time: 11:38).

You said matrix or scalar. It is.

Ask R raise to d 1. What is the quantity on the left hand side? Vector? Scalar? Matrix? Vector; even though it has two indices it is a vector. What is this quantity capturing? At time step t what is the importance of the jth input that is a I will keep asking till everyone replies that is a?

Student: Scalar.

Scalar, ok. Now, you have scalar equal to something multiplied by R^{d1}. So, why do you need this something?

Student: (Refer Time: 12:16).

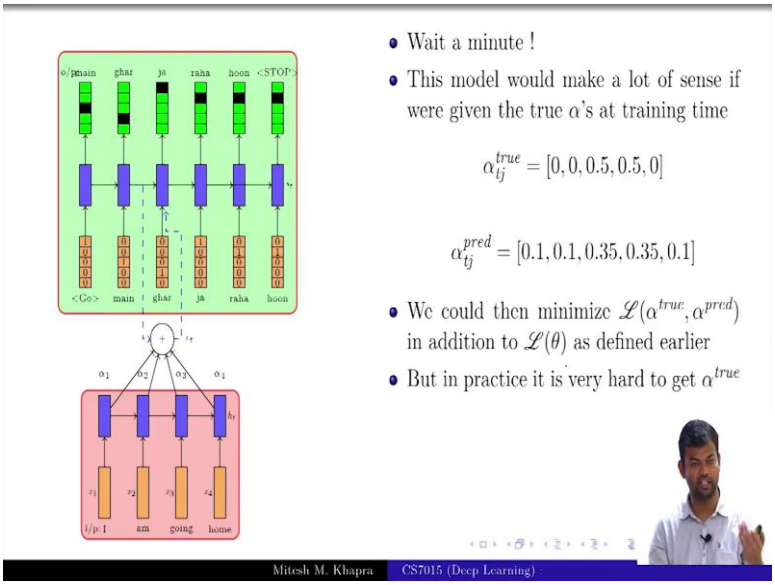
So, what is the dimension of that going to be?

Student: R^{d1}.

R^{d1}. So, that is the dot product. So, you see why we have these parameters, ok. So, what we have done is made it dependent on s_{t1} and h_j, and also made sure that the output is a

scalar that is what these 3 parameters are doing, ok. And these parameters will be learned along with all the other parameters of the encoder and the decoder.

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So, now this is all fine. You would actually someone had given me the true alpha j's and I had predicted this alpha j's to that fancy equation which I just showed you and then I added a dash function, softmax that is the [laughter] safest choice in this course. I want to learn something. So, what do I what should I add?

Student: Loss function.

Loss function. What would the loss function be?

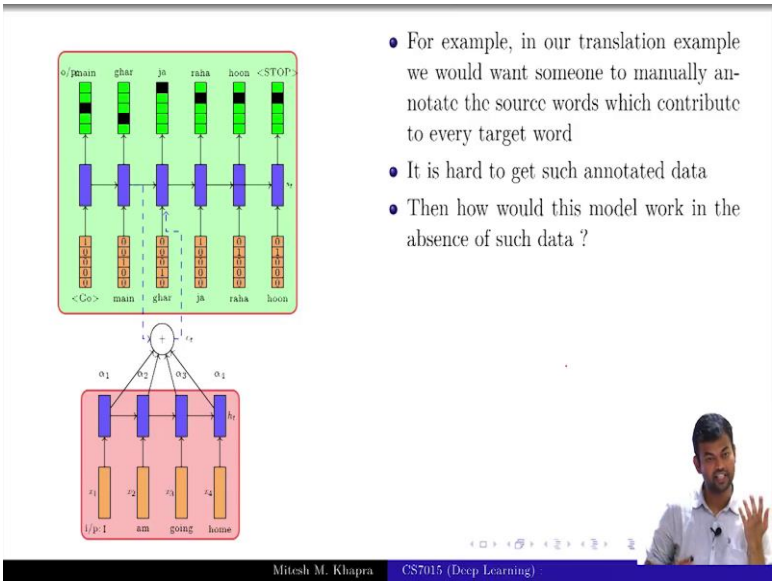
Student: (Refer Time: 13:04).

Say a squared error loss; and then I want to adjust the parameters to minimize this squared error loss. Then all of this makes a lot of sense, right because then you can imagine that your U attention, W attention and V attention will get tuned in a way that the predicted weights are very close to the true weights this we all understand.

Given an objective function we understand that the weights will get adjusted so that you are there to the objective function. But the whole premise was that we do not have the true alphas because in the case of translation no one is going to tell you that the kth word can come came from the jth word or the set of j words. Do you all agree with that? So, if

we had the true alphas this makes a lot of sense because then we could have added a loss function which takes the loss of alpha true with respect to alpha prediction. And then an addition to our lost theta, which was the sum of the cross entropy errors and then we could have jointly minimize this and we could have hope that the attention parameters would have been learnt accordingly.

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In practice we will not have this. In our translation example we would want someone to manually annotate for every word in the output which is the set of input word from is this which it came is not going to be possible this, we cannot collect so much annotated data. So, what do we do? Why should this model then work? They does not have any supervision why should this model work in the absence of such data.

How many of you get the meaning of the question? How many of you see the problem please raise your hands. We are not given the true alphas and that is what a problem is then why should this work better. This works better because this is a better dash, better dash choice language model better dash choice; what is the possibility is there, better modeling choice. Why? Ok, I give you the answer. This was better because this is a better modeling choice. Why so? So, I will give you analogy and the reinforcement learning fans will cringe but they can just go out.

So, suppose I trying to learn a bicycle how to ride a bicycle, that's why I said they will cringe I already see some of you can see as if you guy have a copyright on bicycles, ok.

So, suppose you trying to learn a bicycle, and for some reason you in your infinite wisdom, you decide that you can learn how to ride it without holding the handle, and you start trying to do it. Its conceivable that you know few years or decades you will actually know how to ride the bicycle, right even if you are not holding the handle, right people do that and people can ride it before without that.

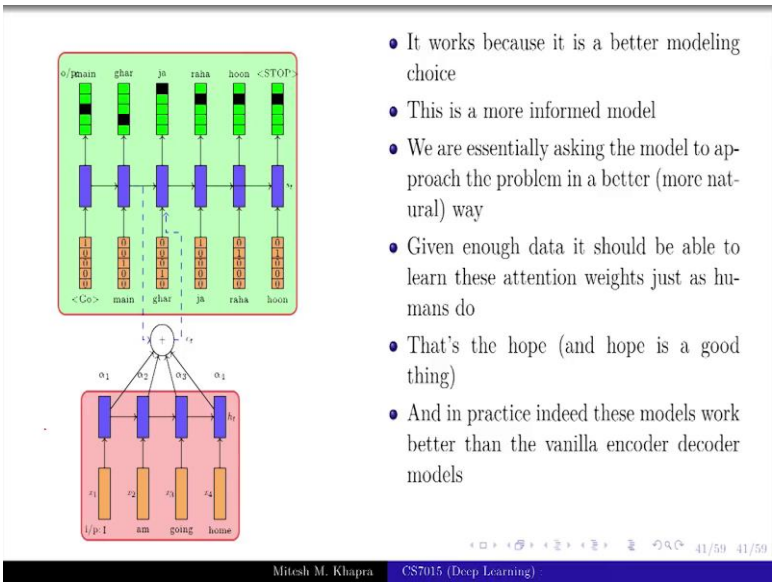
Now, the only thing that I do is I come and tell you that instead of doing this why not you try to hold the handle and then try to ride the bicycle, right that is all I am telling you. I am not giving you any other supervision I have just given you a better model. I have said that instead of just trying to adjust the parameters with respect your feet and the pedal and your back position why not you also introduce this additional parameters where you are holding the bicycle which your hands and now try to figure out what kind of weights you need to put on your left hand, right hand and so on.

I am not giving you any supervision for that that you need to still discover on your own. That you will start riding it you might fall on one side you might fall on the other side. But you will eventually figure out what these weights need to be, right. Because a second model where you hold the handle is a better model than the first model where you are not holding the hand.

In the second model you have additional parameters where you could adjust these parameters. So, that you could learn to drive better that some more natural way more close to human way of learning how to ride a bicycle. The same thing is happening here. The second model where you have a way of learning these attention on the weights even though I am not all I am telling you that look maybe if you decided every time step which were to pay attention on you might be able to do better than feeding the information from the entire sentence at every time step.

That is all the information that I am giving you which is very similar to saying just hold the hand. That is not going to teach you how to ride a bicycle, right. You still have to do the extra work of learning these parameters, but now you are given a chance you are giving the model a chance to learn these parameters they are telling it that, this is a better way of modeling it with this you should be able to learn better, right.

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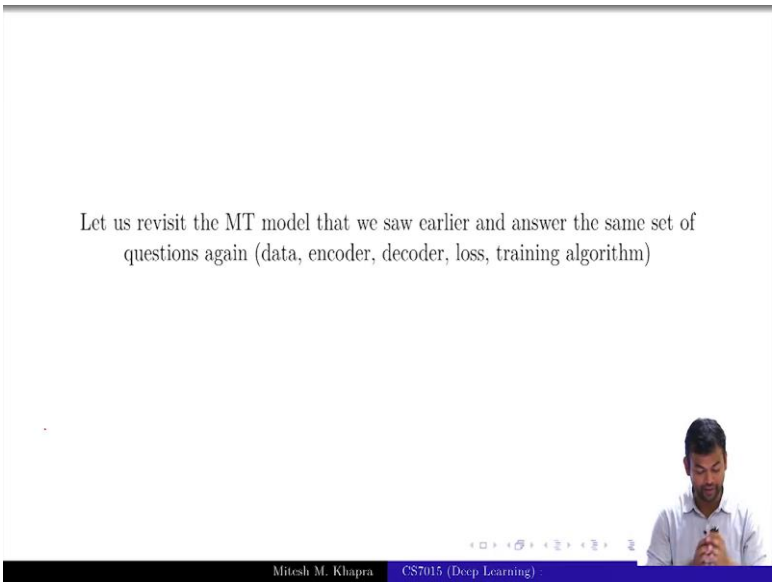


So, there is a hope of doing better because now the model is actually making a more informed choice, right. It is a more informed way of learning how to do translation by focusing on certain words at every time step.

And now these parameters how will they get adjusted? They will get adjusted because at every at a given time step you produced a wrong output you did that maybe because this parameter was wrong which is the v parameter or maybe because these recurrent connections were wrong or maybe because your attention weights are not proper. So, now, adjust the attention weights and that should given sufficient data it should be able to learn which words to focus on, just as humans learn how to do translation, right.

Even when we are doing learning how to translate or when we learn translating from one language to another we are not given this word by word supervision, right. We just do a lot of translations or read a lot of translations and somehow understand that while translating I need to focus on certain words and at every time step this is the word that I need to focus on. So, given enough words it should be able to learn that at least someone gets the joke good. So, that is the hope and in practice indeed these models work better as compared to vanilla encode, you do not know where the statement comes from.

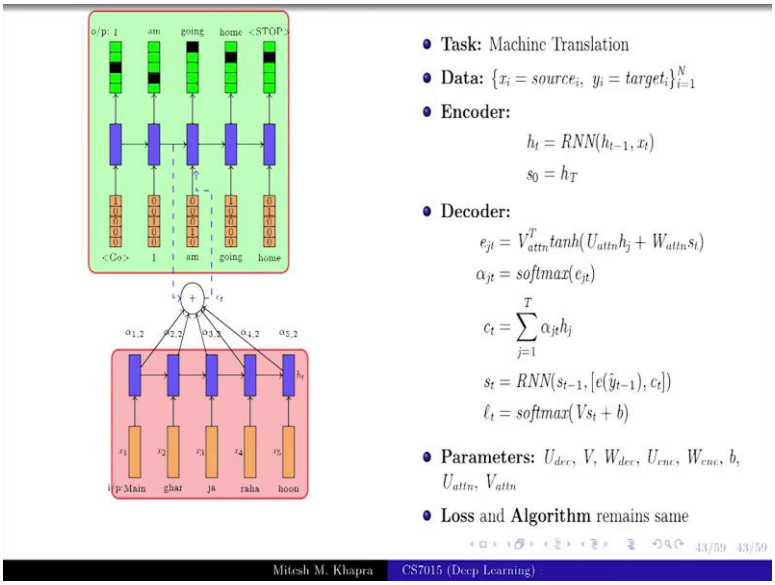
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So, now, let us what we will do is, so this entire thing hints on hope only, right that is all that is all I am saying but it does makes sense, right because you have these additional parameters, which you can learn and you can back propagate through them. I will just not stop there will actually prove what happens not proved by demonstrate what happens in practice, right.

So, with this attention model in mind let us look back at the encoder decoder model that we had for machine translation integrate the attention mechanism with it and then let us see the end to end equation that we get, ok.

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So, this is what the diagram looks like. The input and output still remains the same we have just given the source sentence and the target sentence in particular you are not given which words to pay attention to every time step, right that is not given. So, remember that my input is not changing it still the same source sentence and the target sentence. What is the encoder?

Now, try to work out the equations I wanted to write the equation for y t which is going to be some composite function of x where x is a vector, it is a x 1, x 2, x 3 all the words in the input, and somewhere along the line, it also going to have this attention equation. It is going to take a while but at least try to imagine it there is some hints in the diagram itself you could take a look at it I am just asking you to convert the diagram to a set of equations.

So, encoder part is fine, I have computed the representation of each word at time step t. So, this is a contextual representation the word because it is aware of what the neighboring words are, right. Now, what is the decoder going to be? What is the first thing at time step 1 or a time step t in the decoder, what is the first thing that I am going to compute? The dash weights, the last time step of the decoder of the encoder, sorry.

What is the first thing that we need to compute a time step t? t attention weights. Speak up please. What is the first thing that you need to compute at every time step? How what

kind of a combination I take off the inputs or rather which are the words that I need to focus on from the input. Who tells us this?

Student: Attention weights.

The attention weights. How will you compute the attention weights? Using this fancy equation that we have seen earlier, is this enough I need to convert this to a.

Student: Probability distribution.

Probability distribution, right that is just to ensure that everything is a neat combination. Once I know the attention weights what do I need to feed the decoder? A dash combination of the inputs, a weighted combination. How do I take a weighted combination of the inputs? Summation i is equal to 1 to capital T.

Student: α jt.

α jt into h j, right, no j; t is the decoder time step j is the input word. So, at the tth time step of the decoder I am taking a weighted combination of all my inputs the index over the inputs is going from j equal to 1 to t. By the way did that answer your question that is what you are asking, right, ok, is that fine, ok.

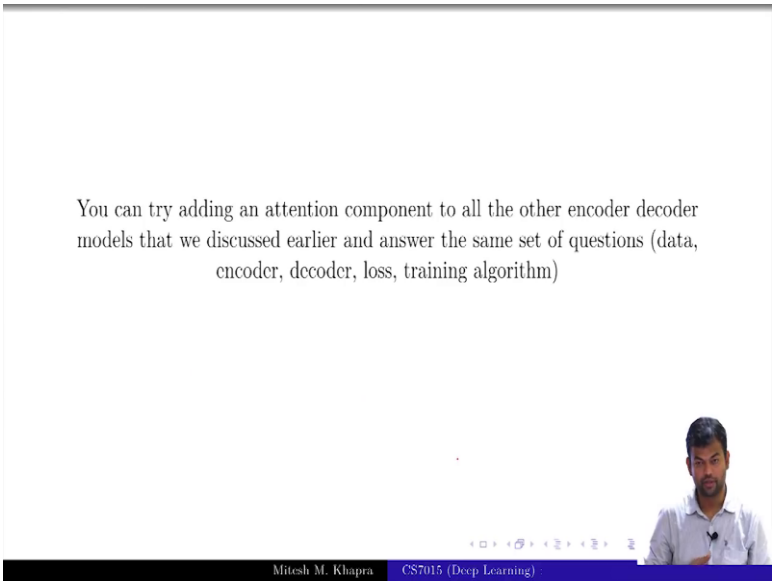
Now, what next now I want to produce a word at the output. What is the decoder going to be? First thing that I am going to do is; decoder is a dash RNN, ok. What is the input to the RNN every time step? The previous predicted word as well as the weighted combination input that you have given it; does that make sense, ok. And then finally, how do I get the probability distribution is that fine yeah I think this should be a distribution, right l t; does not make sense l t is r max of this, right. And what is the loss function?

Student: Cross entropy.

Cross entropy, there is no change in the loss function, right, loss and the algorithm remains the same. Say seen these set of equations. Now, how many of your confident of going back and modifying all the wrong encoder decoder modules that we have covered in the initial part of lecture; modifying them to add the attention equations in it. How

many of you can do that? Please raise your hands, I am not going to ask you just do it so that I feel happy you can do, right. Any questions at this point, very good, ok.

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So, you can go back and try adding attention mechanisms to all the models that we have seen before, right. See how will you compute. So, remember the only purpose of, ok. What kind of a network is the attention network? It is a single feed forward neural network, right, this is just transforming a simple linear transformation of the inputs and in a non-linearity on top of that and then just again one more transformation, right.

It is a simple feed forward neural network. Only these 3 equation somehow need to be fitted in all the other models that we have seen so far, right. This is a very generic framework just as the encoder decoder framework or the very generic framework. The encode attend decode framework is also very generic framework, you can go back and model all the applications that we saw and you can change them change them to at the other case, ok. Try to answer the same set of questions what's the data, what's the encoder, what's the decoder, what's the loss, what's the training function.

And in particular remember that in when you go back represent all the applications that you have done the data is not going to change. No one is going to give us the supervision for the alphas. That is one thing which is not going to change, ok.

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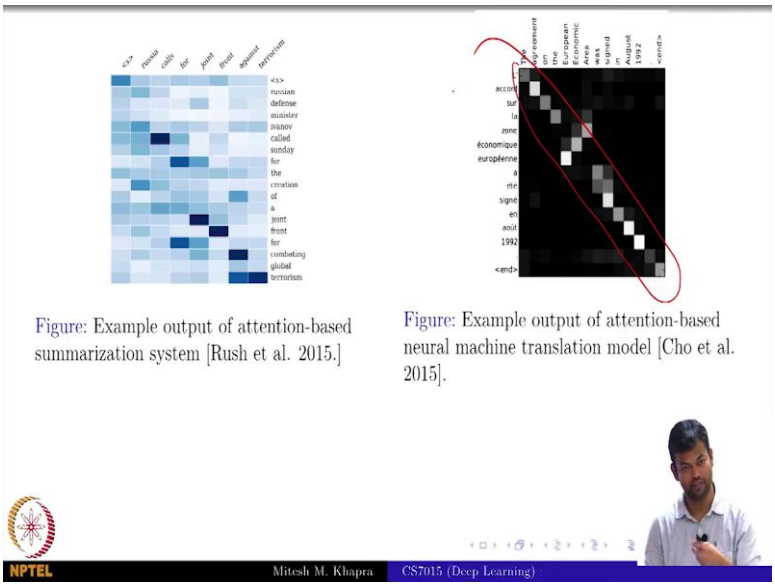


So, here is one more thing. So, this probably tie to this question, like how do we be sure that the alphas actually learn something meaningful. Now, what do I mean by this? If I have to convince you that alphas are actually learning something meaningful, and let us take the context of machine translation. What do I need to show you? Suppose the model has generated an output for a given input sentence it has generated a translation. What do I need to show you to convince that it is learn some kind of weights? At every time step what should I show you?

Student: (Refer Time: 24:46).

What does the attention weights look like, right? So, let us see.

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This is a common trick or not a trick actually it is a probably a trick only, but this is the common thing which is used in several papers and that is why I call it a trick because it is a trick to get a paper accepted that you actually show what the attention weights actually look like, right. So, on this is the input document and this is the summary that you want to generate, ok.

And what you see here is that at different time step, so look at the last time step terrorism it paid maximum attention to the word terrorism in the input, right. So, we can draw this matrix suppose you had capital L time steps in the output and capital T time steps in the input. So, you could draw this L cross T time step or T matrix which tells you what was the attention paid to every input word at every output time step. Do you get that?

You see what is matrix is this heat map is essentially a matrix of size L cross T and every cell here tells you how much attention you paid to a particular word at that time step and the darker the cell that means, more the attention that you paid everyone get this, ok. So, what this is saying is that probably see when you wanted to generate Russia, the maximum attention was paid to Russian and maybe some other words also, sometimes it does not work very well but sometimes it does, right.

So, for calls the maximum attention was paid to called, and then similarly for front with the maximum attention was paid to front and so on. You see some meaningful patterns that it is learning here.

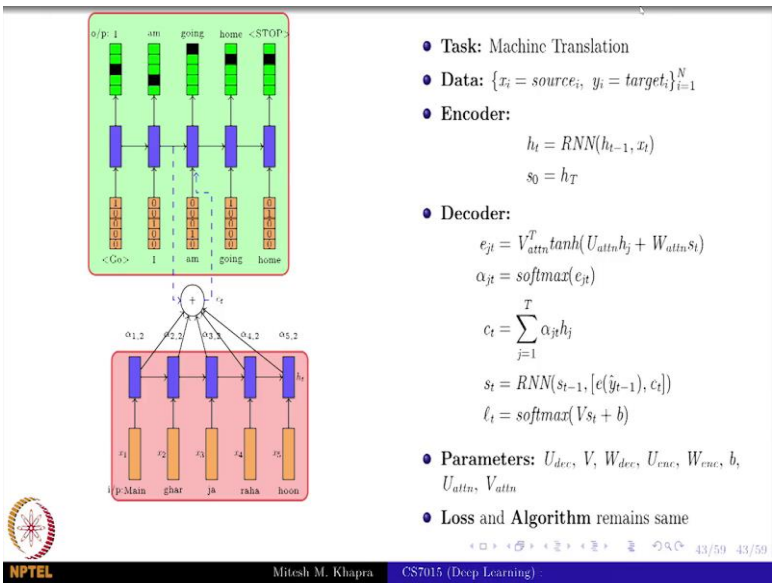
And here is another example for machine translation. So, roughly to quickly understand what this figure is, right. So, this is I think English to French, is it French? Yeah, its French or French to English translation which is largely monotonic, right. That means, at the 4th English word you would end up paying attention to the 4th French word, right that means, you are almost doing a word by word translation.

And that is exactly what you see the most of the attention is along the diagonal, right. So, it is learning some meaningful attention weights. As always helpful if you are using if you are using an attention mechanism to plot the sense see if it is actually learning any meaningful attentions or attention weights or not, right. So, that is a common trick which people use.

Deep Learning
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Lecture – 115
Attention Mechanism

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So, let us start. S last lecture we are looking at encoder decoder models and we saw that a bunch of problems from different domains and different modalities images, text, videos and so on. And, even this cross modal or multi modal applications where you are taking a video and trying to describe it; so video is one modality, description texts is another modality and so on.

We were able to propose modals for all of these using this encoder – decoder architecture. And, then we motivated this attention mechanism where we said that encoder decoder is trying to do this silly thing where it tries to encode the entire input once and that is what how humans do it. He do this back and forth thing where at every time step if we are trying to produce a translation or a single word in the translation we just focus on certain words in the input sentence and kind of ignore the other.

So, the attention mechanism which is this bunch of equations that you see here that allowed you a neural way of modelling attention and the key thing to note here is a there was a supervision for the attention. No one actually tells us that this is the portion of the

text which is important at time step t, but they still works better because this is the better modelling choice and I give you that bicycle analogy and also it is a better modelling choice we are able to no one has given you these supervisions but, you are still have more parameters in the model to learn this kind of a behaviour.

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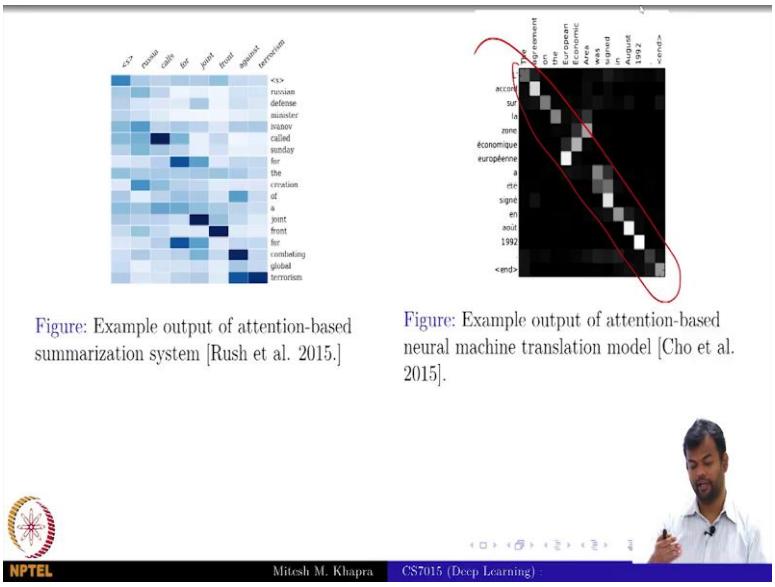


Figure: Example output of attention-based summarization system [Rush et al. 2015.]

Figure: Example output of attention-based neural machine translation model [Cho et al. 2015].

And, then we also saw that we could actually visualise these attention based and from some experiments on some papers we saw that actually learn some meaningful attentions. In the particular case, on the figure on the on the right hand side: so, the one that clearly shows that for a monotonic kind of a translation scenario between English and French. Most of the attentions weights are along a diagram and that is exactly what you would expect, right.

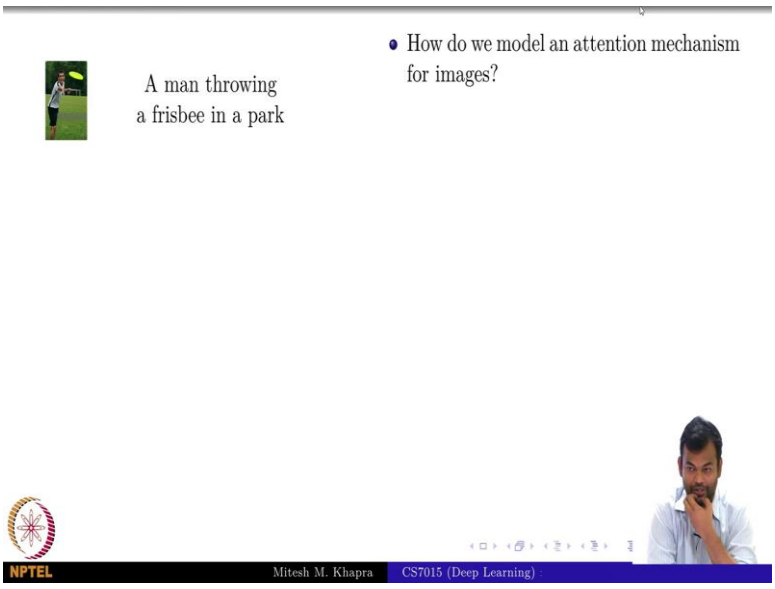
So, that is where we end it.

Deep Learning
Prof. Mitesh M. Khapra
Department of Computer Science and Engineering
Indian Institute of Technology, Madras

Lecture - 116
Attention over images

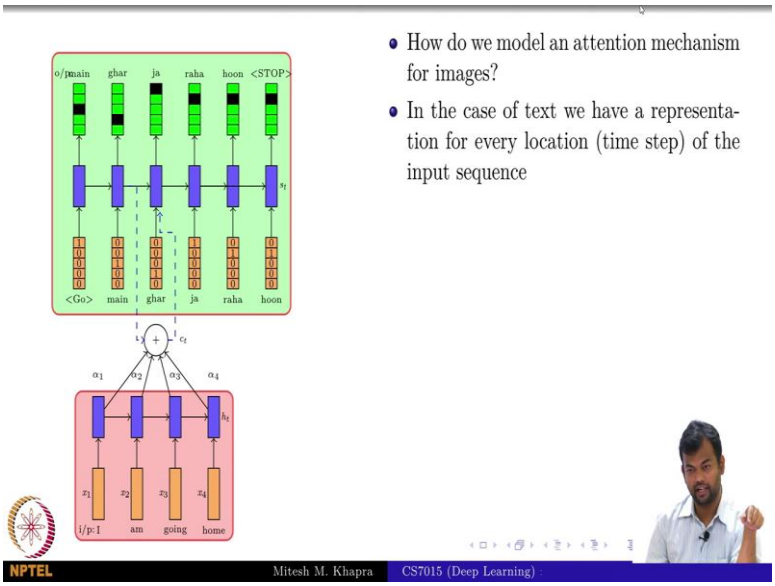
And so now, in this lecture we will go on to the next module which is talking about Attention over images. So, let us first motivate why is it so different and what could be done there right.

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So, the question is how do we model an attention mechanism for images?

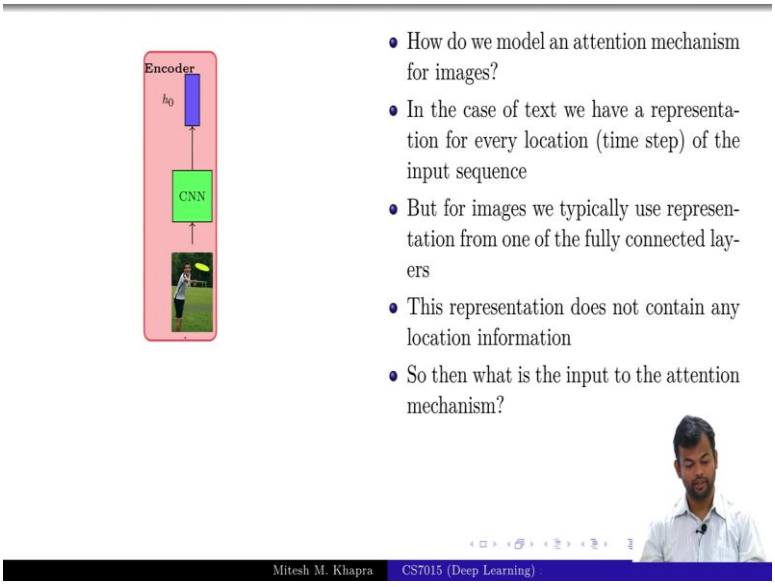
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So, in the case of text we have a representation for every location of the input sequence right. So, every location in the input sequence in the case of text was a word, and then you are looking at this problem of transliteration every location was a character. And whether it is a character or a word, for everything it was discrete.

So, we could just know that this is the important time step t, and then we know that along all the inputs at different time steps you want to pay attention to certain time steps right. So, that the definition they was very straightforward right.

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Now for images what do we do? So, for images we typically take the representation from a CNN right, it could be FC 7, or any of the convolution layers or max pooling layers right. Now, there is no concept of time step there right because the entire image is given to you at one go.

So, now how do you decide where to pay attention to? But if you think about it does makes sense at least the motivation is very clear. So, for example, for this figure, if I am trying to generate the description as man throwing a Frisbee in a garden or in a park or something like that, right.

So, when I am generating a word man, I would want to focus only on the man and not focus on any other part of the image. Similarly, when I am generating the word Frisbee I would like to focus on this. May be when I am saying throwing I would like to focus on his hand action or something like that. And in a park I would like to focus on the background and so on. So, it does make sense that each word in the description is complete covering from coming from a different space in the image, or different position in the image.

But the representation that we use say the FC 7 representation that doesnot contain any location information it just a flat, and vector that we had. So, now how do we do this? How do we get attention on locations? Is is a motivation and the problem clear? And motivation is straight forward. The problem is that we are using FC 7 representation is just a flat vector, remember that was the fully connected vector and does not have any location based encoding.

So, if for example if the fully connected vector is of size 512 I cannot say that the first 12 or first 24 of these 512 dimensions correspond to this set of pixels, the next 24 corresponds to this set of pixels and so on right, so that is the problem. How do I? What do I attend to? How do I decide where to attend to? Because that is what I am saying that the vector the elements of the vector, or the dimension of the vectors do not have any semantic right that is not that the first dimension corresponds to first location.

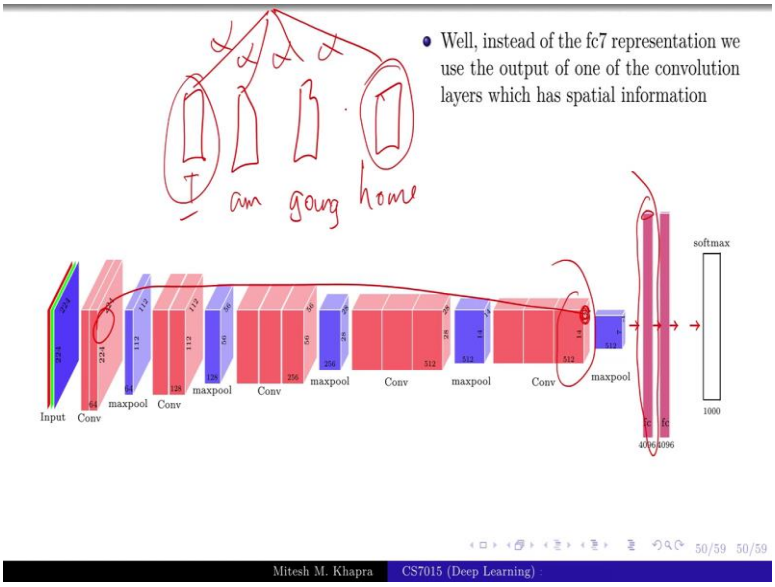
What you want an attention on this? Locations in the image right, but the first dimension in the vector FC 7 vector does not correspond to any specific location in the image you know that was a fully connected vector right. So, it corresponds to everything in the image.

So, what something simpler than that? Why do I say something simpler than that? Object detection is itself is a in itself is a another convolution neural network which does this and so on right. And we saw this past or seen in past class seen in problems.

Now, let us solve the problem at hand. The problem at hand is that I want I will just rephrase a problem definition, so that the answer becomes obvious. I want a representation which allows to give which allows me to get some location information. No, but the fully connected layer if you back track it was fully connected by definition right, the answer is really straight forward. The problem only arise at the fully connected layer right, because that is fully connected. But what about the outputs on the convolution layers? Do they have position information?

Student: Yeah sir yes.

(Refer Slide Time: 03:56)



We saw that suppose this is VGG whatever it is 16 I guess. And this is what I am saying as a problem because this was fully connected. So, you do not know that this dimension corresponds to location 1, location 2 and so on. But if you look at the convolution layers we know that everything here actually goes back to some location in the image. And if I learn to pay attention to this guy maybe I am paying attention to some equivalent portion in the image does that make sense, right?

Now, can you build on this intuition and tell me I want this as a solution right. So, in the case of words these are the word vectors that I had at every location and I was learning to pay attention to them, let us learning these alphas for each of these. Now what is the corresponding diagram for images? What is each of these box is going to be? So, that we learn the attention weights. What do you mean by pass?

Student: (Refer Time: 04:49).

No. So, maybe I am not understanding your answer. What is the equivalent of this, this box which I have highlighted there, between the image and some attention weight? So, you are directly going to operate on the image. Remember attention weights are never given to us no one will mark that man is this, Frisbee is this, and park is this, or whatever. There is no supervision, same size as a convolution what. Now what is the size of a convolution?

Let us take the last one right. So, what do you mean by size? You mean 512, or you mean 14, or you mean the other 14. That is the size of the output 512 cross 14 cross 14. So, what do you mean by the size? 512 or 14, or 14 or 512 cross 14 cross 14. Channel does not make sense because channels capture I mean you do not want to focus on the red part, or the green part, or the blue part in some cases you might that is correct ok.

Student: (Refer Time: 05:39).

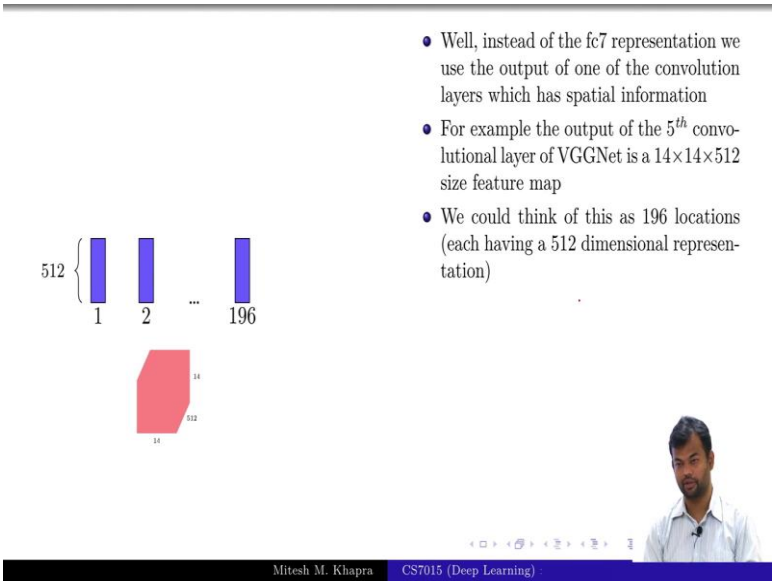
These are all partially correct answers going in the right direction. Let just think a bit more and see. So, probably between these two they gave some part of the answer. Now can you think of it? So, you want a representation for the image first of all of us are clear that we do not want to work with the raw image right that is all of us are clear with that. The second part is we are going to work with the convolution neural network. We want to pick up a representation for the convolution neural network which gives us location information right.

And we agree that the fully connected layer does not give us the convolution layers give this ok. Now, I am asking you to focus on one of the convolution layers which is 512 cross 14 cross 14. So, let us see from there how we will be try to get these attentions ok? So, the output of the 5th convolution layer or 5 c this is I think this whole thing is 5 and

this is 5 a, 5 b and 5 c right. These this is how the code or the general architecture is numbered.

So, this guy has 14 cross 14 locations right, the 512 cross 14 cross 14 output. So, it is 512 channels, but the number of locations is 14 cross 14. And we have seen that each of these 14 cross 14 locations corresponds to certain portion in the image right.

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Now, for each of these 14 cross 14 locations; that means, I have 196 such locations. And for each of this how many dimensional representation do I have? I want everyone to say this.

Student: 512.

512. Because we are reading it from the figure? No. What you are taking is the one taking this pixel, can you see what I am highlighting? I am taking it across the depth right. So, that is why 512 dimensional representation of one pixel in your output volume. And how many such pixels do you have?

Student: 512.

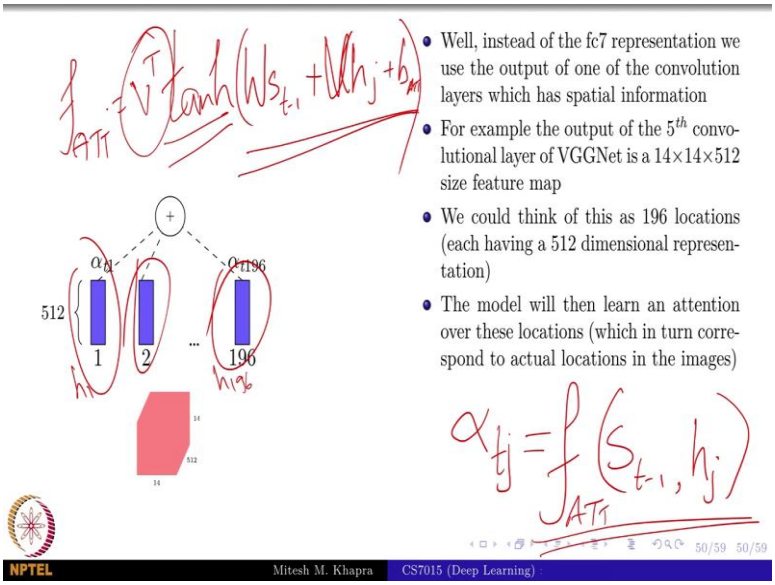
196. And each of these pixels corresponds to some real location in your image; that means, it has space information right ok. So, now these are the 196 locations that you have. Now this looks very much similar to that diagram that we had for words. So, now

you can think about that you have 196 items in your sequence. And now what will you do? We will learn to pay?

Student: (Refer Time: 07:49).

So what would that look like?

(Refer Slide Time: 07:52)



So, at every time step we will have an equation for alphas right. Let us try to write an abstract equation right first of all give me the indices of alpha. What does alpha compute? The importance of the of the jth location at the pth time step fine is that ok? What should this be a function of? Second part is kind of obvious. So, let us call these as h_1 to h_{196}, so these are the j's or the t's j's or the t's?

Student: j's.

j's. So, what would the second parameter be?

Student: h j.

h j. And what is the first parameter be? What is the decoder in this case? We are trying to generate a caption; that means we are trying to generate a sequence; that means, what will be the decoder be?

Student: RNN.

RNN. So what should I depend on s_t or s_{t-1}?

Student: s_{t-1}.

s_{t-1}, s t is the current thing right that we do not know yet. So, it will depend on st minus 1, comma h j. Of course, you can make it depended on several other things also, but at the minimum you will see these two things right because you are trying to understand the importance of these guys, so these better participate in the function. And you are trying to compute the importance at current time step. So, you better know what has happened till time step t - 1, it is not very different from the attention equation that we had written in fact, it is a same actually.

And what was one form of this attention that we had seen? Does anyone remember that? We had carefully analysed the parameters and the dimensions of that form. What should the output of this function be scalar, vector, matrix? Scalar right.

So, what is the form that we had seen for this function? V^T tan h of something. You should get comfortable in writing these equations right because that is what you will do if you are proposing your models and so on right. So, you will see okay, in the previous model this depended on the following two quantities I think it should depend on four more quantities. So, I will write a new equation. Just think about it there are 2 inputs s t minus 1 and h j. What will be doing with each of these inputs? Do a introduce some.

Student: Parameters.

Parameters. So, what will you do?

Student: W (s_{t-1}).

W (s_{t-1}).plus.

Student: Plus U.

V into sorry U into.

Student: h j.

Plus some bias right. So, you just comfortable with this right I mean that is all I mean whatever we have seen. So, first of all remember that the attention is a feed forward

neural network. The moment I tell you that you should know that it will have a linear transformation followed by a non-linearity right. So, that should have been very clear that it would have a linear transformation followed by a non-linearity ok.

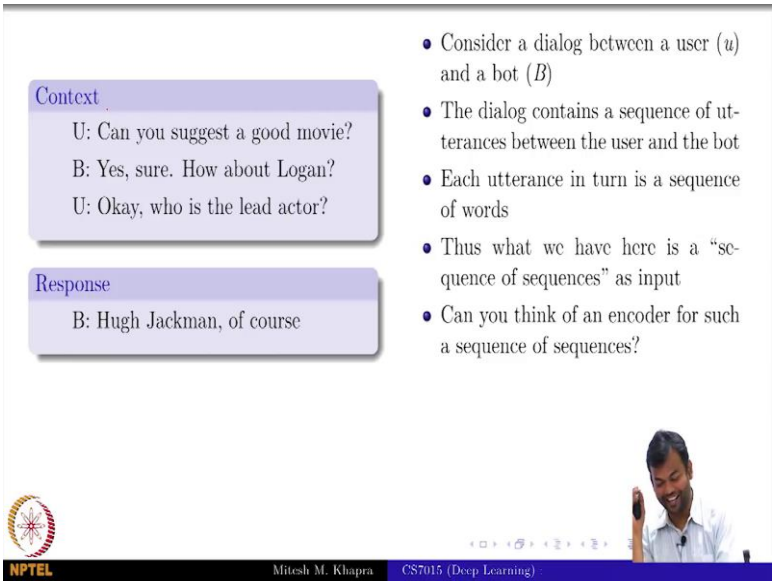
And this is the non-linearity and then you have this other constraint that you want the output to be a scalar. That is why you had this guy which was a vector multiplied by a vector which gives you a scalar. Everyone is comfortable with this, right? So, we see at the attention over images is not very different from attention over sequences. It is more or less the same once you figure out what is the correct representation to you so that you get the space information after that it is straightforward right ok. So, that ends the module.

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Lecture - 117
Hierarchical Attention

So, we will go on to the next one which is Hierarchical Attention. So, again something very popular in nowadays become very common for various things, so again not very difficult idea to understand.

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So, let us first look at the motivation for this and then we will look at the solution. So, consider a dialog right. Today everyone is interested building chatbots; every second start up wants to build their own chatbot. And every second startup out of that they wants to build for the agriculture domain, or the banking domain, or the healthcare domain. So, here is what a typical dialog looks like right? This is of course, not for any profound purpose this is, but you can see this is an important dialog right very relevant and very important. So, this is what a dialog looks like.

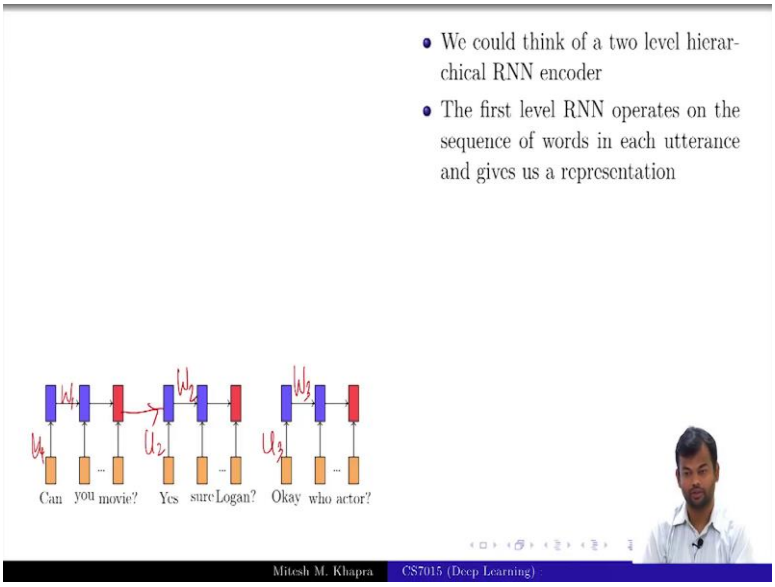
So, let us try to break it down into the kind of entities that we deal with. So, can you tell me about a dialog, what is a dialog? It is a dash; think in terms of things that we have discussed so far.

Student: (Refer Time: 01:07).

Sequence, good right. Again the safest answer is sequence from now on. No, it is only for one lecture. It is a; is it a just a sequence or sequence of sequences right?

So, it contains a sequence of utterances, so each of these lines here is an utterance and each utterance in turn is a sequence of words right ok. So, what we have here is a sequence of sequence as input and this is very common in many many applications right. So, can you think of an encoder for such a sequence of sequence? RNN of RNN's good that is the answer right.

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So, we think of a two level hierarchical RNN encoder. So, first leveller will encode the utterances ok. Let me ask you few questions. Is there is a mistake in a diagram? Should this be connected? Yes, no, maybe, do not care ok. Second question; I will write some parameters here right, what is our notation; w, u. This is u, right and this is w right. Is it fine? If I have a dialog which contains 100 utterances, what will happen? That is a practical problem.

But more conceptually what is wrong here? What is each RNN trying to do? I encode a sentence, encode an utterance. So, why should it be different for the first utterance, second utterance, third utterance and so on right. All these RNN's should be the same; does that make sense? The U 1 is equal to U 2 is equal to U 3 and w 1 is equal to w 2 is

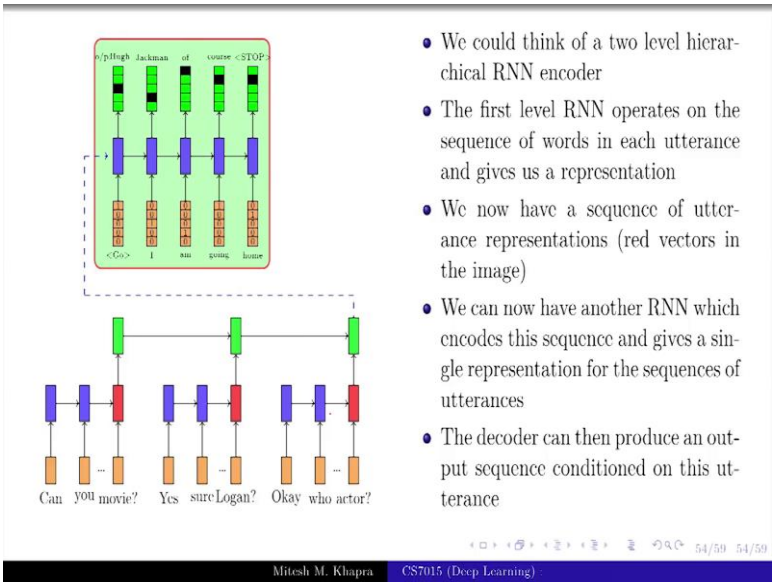
equal to w 3 ok. Is that fine, everyone agrees with that? So, now can you tell me if there is a correction should be there, or not?

Conceptually what is each RNN doing? Encoding?

Student: (Refer Time: 02:47).

One sentence right, then why should it be connected to the previous sentence? But then if you do not know all these sentences then how will you predict the utterance, the next response rather, what is missing here? What kind of a what was the title of this module? So, what is missing? Where is the hierarchy right?

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So, what we will have is this right. So, each of these green guys is presentation for one utterance in your input. In fact the, red red guys sorry. The red guys are the representations for the utterances in your dialog and then the green guy is a sequence of utterances right. So, remember you have the sequence of sequence of words.

So, the red guys are the sequence of words, and the green guys are the sequence of utterances. Does that make sense ok? How many of you get this? Please raise your hands. Ok, good. So, and now what would the decoder be? You have a hierarchical encoder. What could the decoder be? What is the decoder have to do? It has to produce a everyone?

Student: Sequence of words.

Sequence of words. So, what kind of a decoder will you use? Just an RNN not a hierarchical RNN, the input is hierarchical why should not the output be hierarchical? How it would look unbalanced right? The diagram will not look very neat right if you (Refer Time: 03:57).

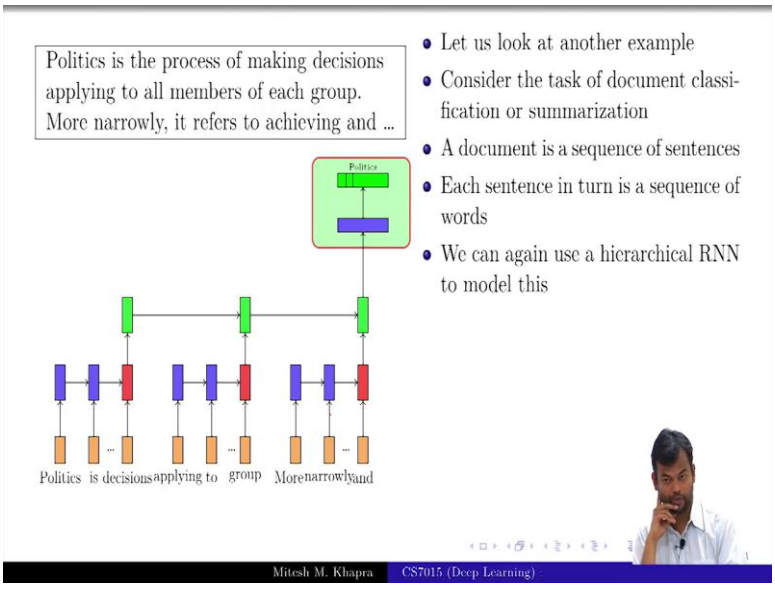
Do you need a hierarchical and decoder? No right, I mean just a simple decoder. Because the decoders has to produce a sequence of words, so it will take something from the encoder, what will it take from the encoder in the normal encoder decoder paradigm not the attention based paradigm? What will be the input to the decoder?

The last dash vector, your options are state that is very safe answer here, the last state vector what will that means? Your options are orange, blue, green and red.

Student: Green.

Green; the last green vector that is what the input is going to be, right. So, this is what is going to look like. This is option 1. What is option 2? Feed it to every time step that is what option 2 is going to be ok. So, that is you have your hierarchical encoder decoder network.

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What is missing here? Attention, ok. So let us look at another example. Consider the task of document classification, or summarisation. What is the difference between two? In classification what the what were the decoder be? Feed forward neural network with the softmax. What would be decoder be in the case of summarisation?

Students: RNN.

RNN good. What is the document? Sequence of sequence, not sequence of sequence of sequence. It can be a sequence of sequence of sequence also right. I did not think of that then it could be a sequence of sequence of sequence of sequence ok. Let us look at the not so funny case which is sequence of sequence. What is the sequence of sequence of?

It is the sequence of sentences, which in turn is a sequence of words, which in turn is a sequence of character. We will not go there; we will just keep it till words. So it is a sequence of sequence. So, again you need some kind of a hierarchical encoder here right. So, will encode each sentence, then you will treat the sentence sequence as a sequence, encode that and then you will pass it to the classifier.

Now think of this problem right. Now if we want to do document classification, how would you go about it actually? You want to classify whether this is a politics or sports or health or whatever (Refer Time: 05:57). I think in terms of attention, what would you do actually? First we will find the important words in the sentence. To find the important words you will have to read all the.

Students: (Refer Time: 06:10).

If you want to find the important words you will have to read all the sentences. So, what will you do? First find the?

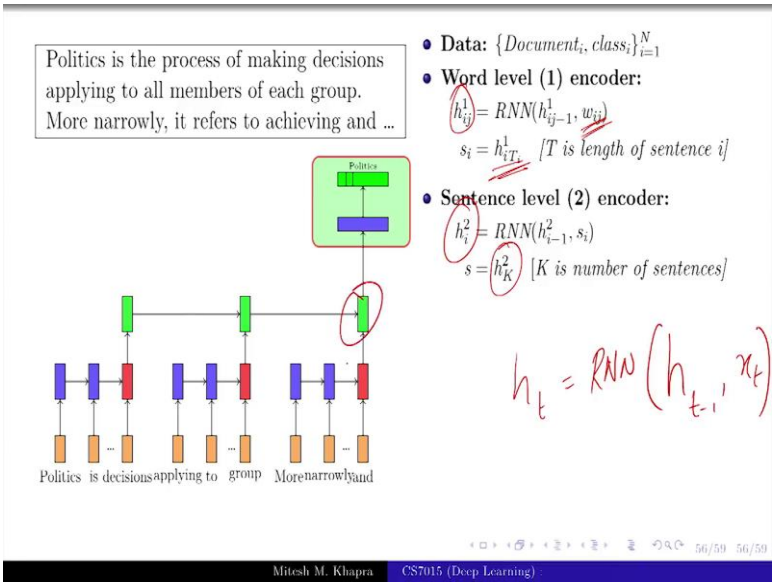
Students: Word in sentences.

Word in sentences and then? Important words within the sentence. So, what kind of an attention mechanism do you need?

Student: Hierarchy.

Hierarchical, right.

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So, let us look at this; so first let us look at the our data model paradigm. So, what is the data given to you? I given a document and the class table for the document. And your first thing is the word level encoder which looks like this, can I will not explain what this is I will just expect you to know what this is. Why do I have two indices for h, what is i, and what is j? i is the dash id, sentence id and so it is the jth word or the ith sentence right that is what I am encoding. And how many sentences am I encoding? The number of sentences in the document ok.

And what is the second encoder? So, diagram is absolutely clear, but the equations are not; the diagram is absolutely clear right. There is no just need to map the red, blue, orange, green, guys to the equations ok. So, let me ask you this, what is h_{ij}? No, in the diagram? Blue, red, orange?

Student: (Refer Time: 07:23).

What is w_{ij}? How did you write the RNN equation? Time step t is equal to RNN of t minus 1 and the input at time step t right. What is the input at time step t here?

Student: Word.

Word, right. Probably this was not a good choice maybe we should make it x_{ij}; w, I think we might get a confused with the weight's we should not, but they are. So, fine, so w_{ij} is

actually the input word, what is h i j minus 1? Now tell me what colour is w i j? It is like an IQ test at which colour map it, w i j maps to which colour?

Students: Orange.

Orange, good and h i j?

Students: Blue.

Blue, but what about the red? That is which colour I mean sorry not which variable.

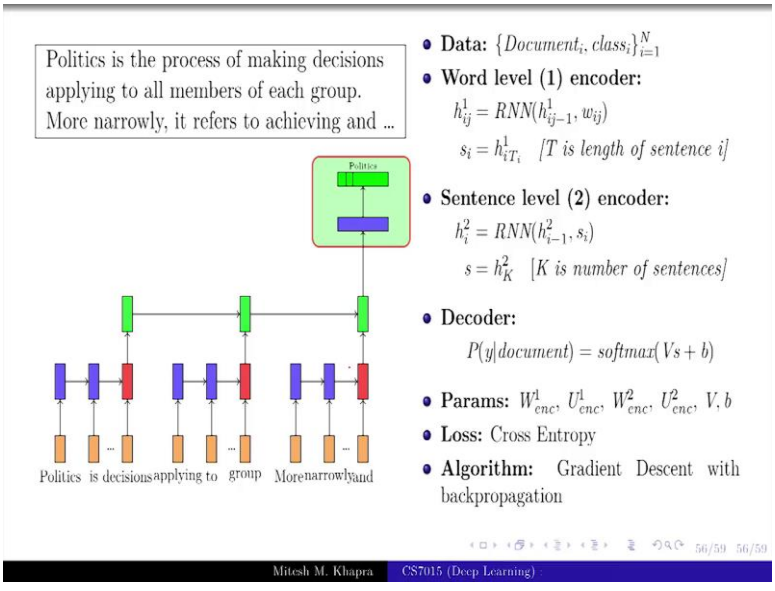
Students: (Refer Time: 08:15).

hi Ti that is the last state of every sequence right. T i is a length of the ith sentence right ok. Now, what about h_{i2}? The green guys right and what is this h_{2k}?

Students: (Refer Time: 08:29).

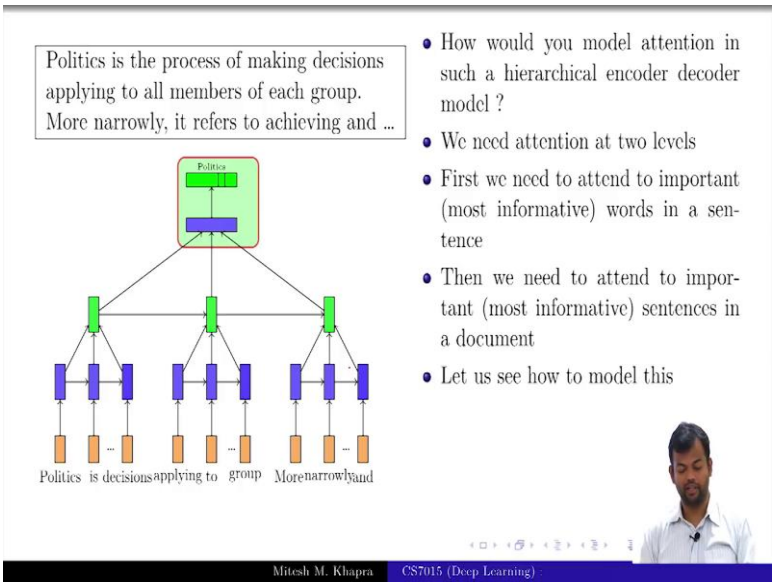
The last green guy, this guy, ok, is it fine?

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And then the decoder is just a softmax, we do not need to go with that and loss and everything is fine. So, this again whatever it is we should always be comfortable and writing the end to end equations from x to y right and you can write it in this case.

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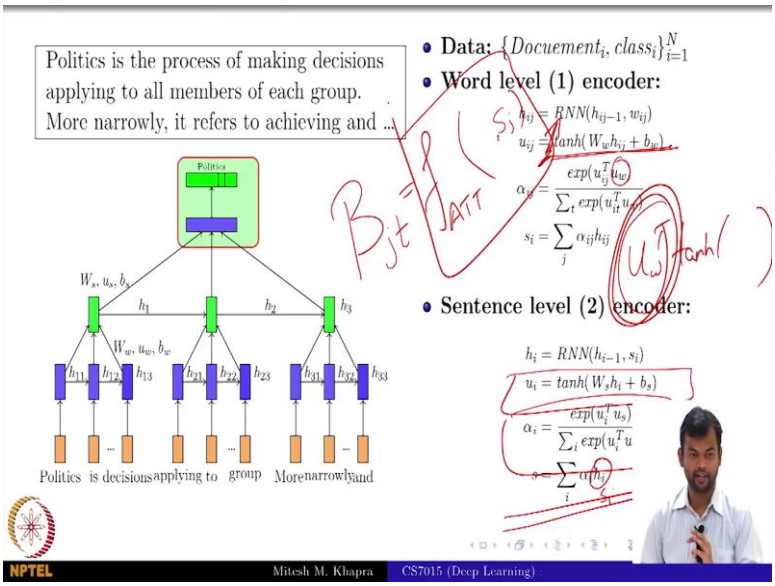
Now, let us make it a bit interesting. How would you model attention in such a hierarchical encoder decoder model? How many attention functions would you need? Two; one for attention over sentences, the other for attention over?

Students: Works.

Works ok. Can you think of these equations? Not a very big stretch from what we have done already right. I mean at level 1 it should be straight forward, at level 2 just ignore level 1.

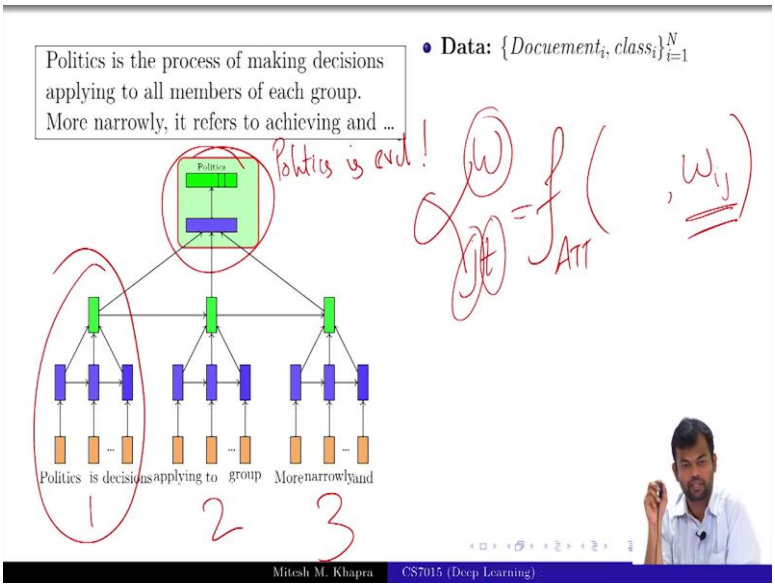
How many if you can imagine the equations? It is not very hard I am not joking I am I mean just think about it. And the level 1 should be straight forward because that is just the same as ok. So first we need to attend to the most important words in a sentence and then we need to attend to the most important sentence in a document ok.

(Refer Slide Time: 09:35)



Let us have see how to model this. So, we have document, again the same input, then you have the word level RNN ok. Now what be the word level attention equation look like?

(Refer Slide Time: 09:49)



I am looking for the attention equation for words. What are the indices going to be? J, i j t, ok. What is i j t? That is I have put as superscript in w. This is the word level attention. So, at the tth time step I want the importance of the?

Student: (Refer Time: 10:04).

j^{th} good, right. What would that equation depend on?

Students: The word.

The word should be straightforward right, it should depend on oh but oh ok sorry. This is only for this guy right, ok. So, you have focusing on one of these; so you trying to find the importance of these three words right which are there in the first sentence.

So, you have computed h i j; that means, you have computed all the representation for each of these word. That means, you have computed the first three blue vectors that you see ok. And then you are computing the attention, no, so this is, oh, so instead of having it here. This is how we have been writing at right.

Student: (Refer Time: 10:49).

Ok, I so sorry I should have check this. So, read this as u i j is equal to, or let me just explain it. So, remember this is a vector and we wanted to do this operation to make it a scalar right. So, u w is that parameter which was getting multiplied earlier. So, we had this attention equation as u w transpose tan h of something right. So, now, that u w has been removed from here in equation 2 and has been added as exponent to equation to the alpha equation. Does is that ok?

How many of you completely confused? Please raise your hands. How many of you understand this completely? Once can the sum the 1, ok, aa shit, what did I do? Ok let us just see if I can still salvage this. Ok let us go one by one. So, what is let me just delete some of these things, let us try to write it on our own right. So, this is what we are trying to do.

So, I will I am ignoring the sentence id rights. So this is sentence 1, 2 and 3; so let us just focus on one sentence and the same equations we hold for the other sentences also ok. So first of all the attention weight would depend on what? It would depend, what are we trying to pay attention to?

Student: Words.

Words; so it should have word in the input right. What else can you at have in the input?

Student: Previous.

Previous state; unfortunately for this problem do we have a previous state?

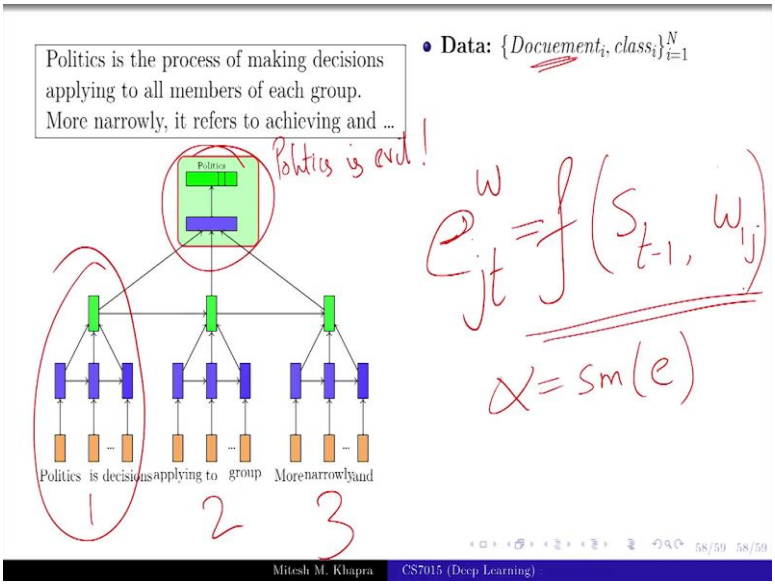
Student: (Refer Time: 12:27).

No, we just doing one prediction right, there is no RNN here, we just the feed forward network. Do you have any s t minus 1 in the output? That we have put here. So, this was the importance of the jth word at the tth time step. So j belongs to the input and t belongs to the decoder right. In this case does decoder have multiple time steps? There is only one time step of the decoder right that is the problem which we have run into.

But let us assume instead of classification we are trying to do summarisation; that means, we are given this document. And we were trying to generate summary of this and let us say the summary was the following ok. So, this is the summary that you are trying to generate from this document ok. And now this summary has 3 or 4 time steps if you count exclamation as the last time step. Now, how is it going to be? What is the decoder going to be in that case? RNN right and the decoder will have some k time steps ok. Now, at every time step at a given time step t, what am I trying to do?

In next assignment try to develop a better eraser for this, ok.

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So, we want to compute, when you compute the attention for a word; the jth word in particular at that t^{th} time step, right. And we have for a minute understood that we do not have a feed forward network at the decoder. We have a recurrent neural network at the

decoder because we are trying to generate a summary, ok, we are trying to generate a sequence at the output. So, at the tth time step we are interested in understanding which of the document was to pay attention to ok. So, now, that is going to be a function of what and I finded a bit irritating for the want of a better word that at least one input to this function should be straight forward right. What is it?

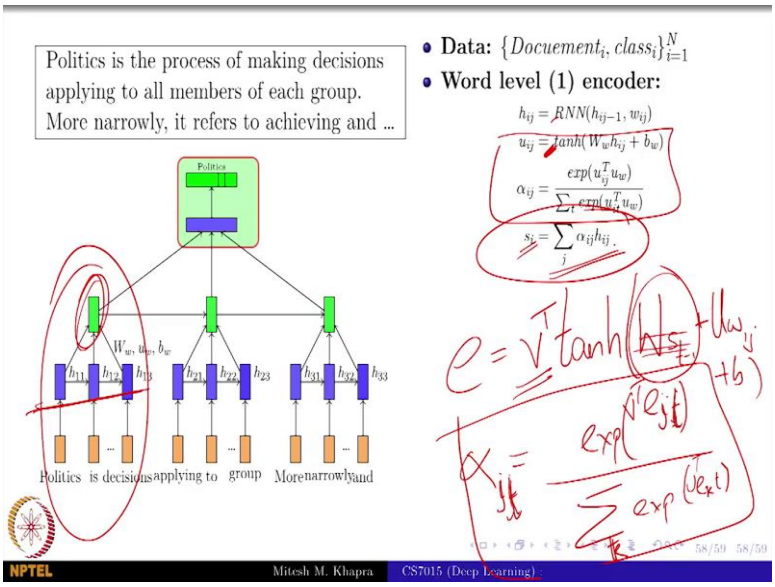
Student: The word.

The word, that you are trying to learn how much attention to pay to right. So, that should be very straightforward, so that should be w 1 j because I am considering the first sentence right now ok. Is that fine? For the first sentence I am trying to find out which are the words which are important. What is the second input that you could put in? It should depend on the index t right, so it should be the previous state of the decoder ok. And then of course, I will have a this is again actually not alpha, but e right. And then you get how do you get alpha from there, how do you get the alphas?

Student: Softmax.

Softmax, is that fine ok. So alpha will be some softmax of the e's; is that ok, fine. So, this is for the word level. Now, the equations that are written on the slide are slightly. So, the equation has slices slightly differently written, so let me just go back and write our own equation. And so we want to write an equation for this, what was our equation?

(Refer Slide Time: 15:26)



Ignore the equations on the slide. it was something like this; v transpose tan h of w.

Student: s t.

s_{t-1} +U_{w1j}+?

Student: b.

b ok. Now, imagine that your decoder is a feed forward neural network. What will be missing in that case? There is no s t minus 1; there is no previous state of a decoder because we just want to make a prediction once by paying attention to all the important words and sentences in your document right. So, which part will go away? ws_{t-1} ok, fine.

Now, the other thing that you are doing is alpha was actually or other alpha i j is actually exponent of e i j divided by summation of other k sorry; alpha j t e j t, e k t; is that fine? It is just the softmax equation; is that ok right? Now, the only thing that you see different and these two equations here is, first you do not have the w s t minus 1 because the decoder only has one time step. And second we have taken out this v transpose from here and instead we have added it here. Is that ok? Does that make sense? This is just different way of writing it. So, again you write the attention equation for the words now ok. Now what about the sentence? Now first of all, earlier what were we using for the green guy, what was the representation for the green vector? It was the; what was the green vector?

In the absence of attention what was the green vector? h t i right, the last time step of sentence 1. Is it fine everyone with me? please raise your hands, if you are with me ok. Now what would it be? It would be a dash sum of w vectors, a weighted sum, attention weighted sum right; so that is exactly what this equation is capturing. So, what did you saying is there is representation of sentence i is a weighted sum of the representations of all the words in that sentence is it ok.

So, that we will get a representation for s 1, s 2 up to s capital K all the sentences that you have. Now what do you want to do for the second level what do we want? We want to compute the importance of that sentence for the tth time step right. So, let us call that beta. So I am interested in beta if we need to really read out this I said again alpha is being used in both the places right.

So, you want to find out the importance of the j-th sentence at the t-th time step. What is it going to be a function of? 1 is a sentence representation, what is the sentence representation given by?

Student: s j.

s j and what else? The decoder state at the previous time step right. Does the decoder have a previous time step state here? No. So, it will just depend on s j and that is exactly what this equation is capturing right. And again the same trick that I have added this extra parameter to the exponent. Is that fine? And the final representation being fed to the feed forward network is a weighted sum of the sentence representations right. So, this again has to be $s_i \alpha s_i$ ok.

I really sorry about this, but I am pretty sure that once we correct the slides and then you go back and look at it should be clear right. It just two sets of equation, one set of equation sorry os sorry that is correct sorry. So, this the idea is there are two sets of attention mechanisms, for each you will have your own set of equations. The basic form if you can work out what the f attention would depend on. The actual form would depend from would differ from paper to paper or the toolbox to toolbox. That does not matter so much, you just need to know that you have these as the input. You are going to add some parameters to every input that means, you are going to do linear transformation. And then you just need to make sure that alphas eventually turn out to be scalars right. That is why you will have this additional vector getting multiplied at one point.